

PAA3905E1-Q: Optical Motion Tracking Chip

General Description

The PAA3905E1 is PixArt Imaging's optical motion tracking chip specifically designed for low light condition operation. The state of the art of architecture allows motion tracking under low light condition as low as 5 lux. The chip supports wide working distance of 80 mm to infinity under challenging conditions to cater different far field application needs and ambient conditions. It is suitable for far field tracking application in providing motion tracking with stable hovering function for Drone.

Key Features

- Wide working distance from 80mm to infinity
- No focal length calibration required
- Auto detection of challenging conditions, e.g. checker boards, stripes, glossy surface and yawing
- 16-bit motion data output with motion detect pin output
- Internal oscillator
- Raw Data Output through register read
- Support synchronized multi-chip operation
- Automatic switching of Operation Mode

Operation modes

Mode	Description	Lux (Typ)
0 @ 126 fps	Bright Mode for general motion tracking	60
1 @ 126 fps (Default)	Low Light Mode for low light motion tracking	30
2 @ 50 fps	Super Low Light Mode for super low light and low speed motion tracking	5

Applications

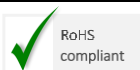
- Drone or other devices that require far field motion detection and hovering stability
- X-Y positioning in GPS denied environment

Key Parameters

Parameter	Value
Supply Voltage	VDD: 1.8 to 2.0 V VDDIO: 1.8 to 3.6 V
Working Range	80 mm to infinity
Interface	4-Wire SPI @ 2MHz
Speed	max 7.4 rad/s (Mode 0 & 1)
Effective Viewing angle	42°
Power Consumption	3.5 mA at RUN state
Package size (L x W x H)	12-pin LGA Package with L242-ZSZ1 Lens 4.0 x 5.0 x 3.0 mm ³

Ordering Information

Part Number	Description	Package Type	Packing Type	MOQ
PAA3905E1-Q	OMT Chip	12 pins LGA	Tube	2000
L242-ZSZ1	Lens	Plastic lens	Tray	2000



For any additional inquiries, please contact us at <http://www.pixart.com>

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1.0 Introduction

1.1 Overview

PAA3905E1-Q contains a Picture Element Acquisition System (PEAS), a hard-coded Digital Signal Processing System (DSPS), and a four-wire serial port interface. It is based on Optical Navigation Technology to measure changes in position where acquiring sequential picture elements for calculating the direction and magnitude of movement (delta X and delta Y).

Note: Throughout this document PAA3905E1-Q is referred as the chip.

1.2 Block Diagram

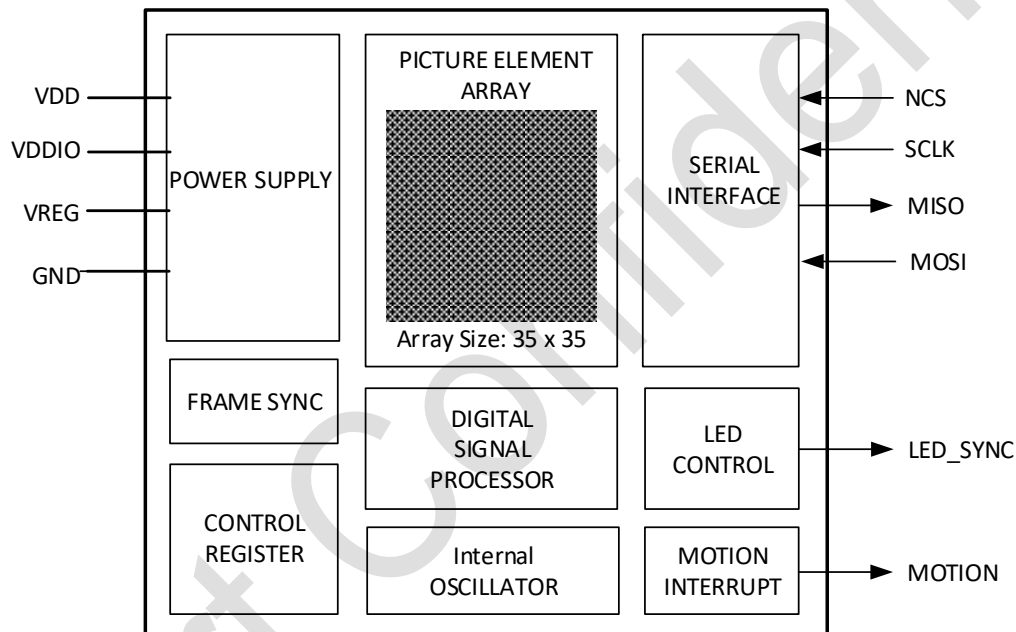


Figure 1. Block Diagram

1.3 Terminology

Term	Description
DSP	Digital Signal Processing System
ESD	Electrostatic Discharge
LED	Light Emitting Diode
IC	Integrated Circuit
I/O	Input / Output
IR	Infrared
PCB	Printed Circuit Board
IMU	Inertial Measurement Unit
AMS	Auto Mode Switching

1.4 Pin Assignment and Signal Description

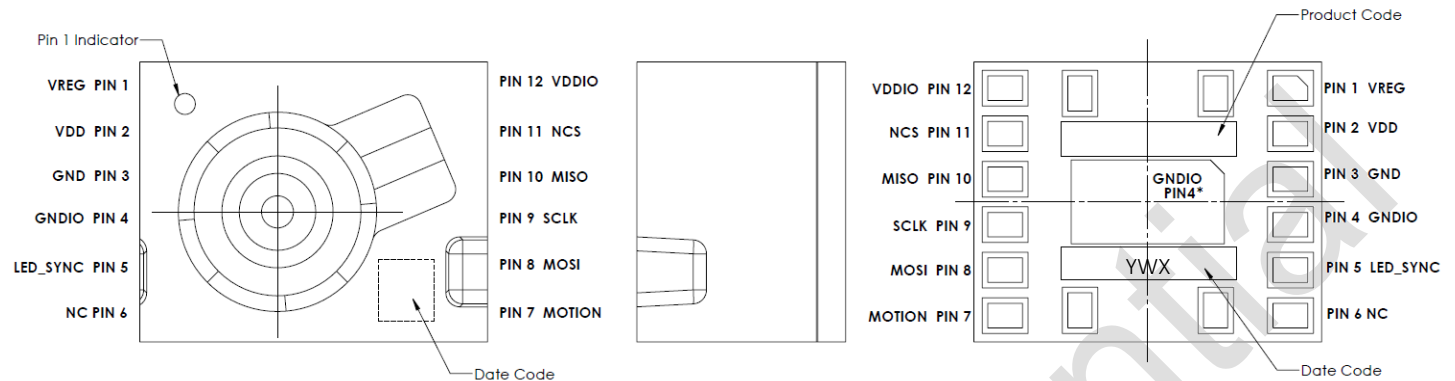


Figure 2. Pin Configuration

Table 1. Signal Pins Description

Function	Pin No.	Signal Name	Type	Description
Power Supplies	2	VDD	Power	Input power supply
	12	VDDIO	Power	I/O reference voltage
	1	VREG	Ground	Internal voltage output
	3	GND	Ground	Ground
	4	GNDIO	Ground	I/O ground
	4*	GNDIO	Ground	I/O ground pad
Control Interface	8	MOSI	Input	Serial data input
	9	SCLK	Input	Serial data clock
	10	MISO	Output	Serial data output
	11	NCS	Input	Chip select
Functional I/O	7	MOTION	Output	Motion interrupt (Active low)
	5	LED_SYNC	Output	External LED control pin (Active low).
Reserved	6	NC	NC	No Connect

2.0 Operating Specifications

2.1 Absolute Maximum Ratings

Table 2. Absolute Maximum Ratings

Parameters	Symbol	Min.	Max.	Unit	Notes
Storage Temperature	T_S	-40	85	°C	
Lead-Free Solder Temperature	T_{SOLDER}		260	°C	
Supply Voltage	VDD	-0.5	2.0	V	
	VDDIO	-0.5	3.6	V	
Input Voltage	V_{IN}	-0.5	3.6	V	All I/O pins
ESD	ESD_{HBM}		2	kV	All pins (Human Body Model)

Notes:

1. The above maximum ratings are the guaranteed parameters that the chip expects to be functional as per intended design.
2. Exposure to these conditions or beyond those indicated may adversely affect device reliability and unexpected or premature functional failure.

2.2 Recommended Operating Conditions

Table 3. Recommended Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit	Notes
Ambient Temperature	T_A	0		60	°C	
Power Supply Voltage	VDD	1.8	1.9	2.0	V	Including supply noise
	VDDIO	1.8	1.9	3.6	V	VDDIO ≥ VDD
Power Supply Rise Time	t_R	0.15		20	ms	0 to VDD min
Supply Noise	V_{NA}			100	mV _{p-p}	10 kHz to 75 MHz
Serial Port Clock Frequency	f_{SCLK}			2	MHz	50% duty cycle
Working Distance	Z	80			mm	
Effective Viewing Angle	V_A		42		°	
Minimum Illuminance ² (@ Crimson Carpet, Grey Vinyl & Light Grey Cement surfaces)	L_{XM0}		60		lux	Mode 0: Bright Mode
	L_{XM1}		30		lux	Mode 1: Low Light Mode (Default Mode)
	L_{XM2}		5		lux	Mode 2: Super Low Light Mode
Frame Rate	F_{RM0}		126		fps	Mode 0: Bright Mode
	F_{RM1}		126		fps	Mode 1: Low Light Mode (Default Mode)
	F_{RM2}		50		fps	Mode 2: Super Low Light Mode
Speed	S			7.4	rad/s	Mode 0 & 1

Notes:

1. PixArt does not guarantee the performance of the system beyond the recommended operating condition limits.
2. In addition to visible light spectrum (lux), the chip is also sensitive to IR spectrum up to 940 nm of wavelength which aid in tracking under low light ambient condition.

2.3 DC Characteristics

Table 4. DC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
Supply Current	I_{DD_RUN}		3.5		mA	Average current. No load on MISO, MOTION.
Power Down Current	I_{PD}		12		μA	
Input Low Voltage	V_{IL}			$0.3 \times V_{DDIO}$	V	SCLK, MOSI, NCS
Input High Voltage	V_{IH}	$0.7 \times V_{DDIO}$			V	SCLK, MOSI, NCS
Input Hysteresis	V_{I_HYS}		100		mV	SCLK, MOSI, NCS
Input Leakage Current	I_{LEAK}		± 1	± 10	μA	$V_{IN} = V_{DDIO}$ or 0V, SCLK, MOSI, NCS
Output Low Voltage	V_{OL}			0.45	V	$I_{OUT} = 1mA$, MISO, MOTION
Output High Voltage	V_{OH}	$V_{DDIO} - 0.45$			V	$I_{OUT} = -1mA$, MISO, MOTION
Transient Supply Current	I_{DDT}			70	mA	Max supply current during the supply ramp from 0V to V_{DD} with min 150 μs and max 20 ms rise time (does not include charging currents for bypass capacitors).
	I_{DDTIO}			70	mA	Max supply current during the supply ramp from 0V to V_{DDIO} with min 150 μs and max 20 ms rise time (does not include charging currents for bypass capacitors).

Note: All the parameters are tested under operating conditions: $V_{DD} = 1.9V$, $V_{DDIO} = 1.9V$, $T_A = 25^\circ C$.

2.4 AC Characteristics

Table 5. AC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
Motion Delay After Reset	$t_{MOT-RST}$	50			ms	From reset to valid motion, assuming motion is present
Shutdown	t_{STDWN}			100	ms	From Shutdown state active to low current
Wake from Shutdown	t_{WAKEUP}	300			ms	From Shutdown state inactive to valid motion.

Note: All the parameters are tested under operating conditions: $V_{DD} = 1.9V$, $V_{DDIO} = 1.9V$, $T_A = 25^\circ C$.

2.5 SPI Specification

Table 6. SPI Timing Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
MISO Rise Time	t_{r-MISO}		50		ns	$C_L = 100\text{pF}$
MISO Fall Time	t_{f-MISO}		50		ns	$C_L = 100\text{pF}$
MISO Delay After SCLK	$t_{DLY-MISO}$			120	ns	From SCLK falling edge to MISO data valid, no load conditions
MISO Hold Time	$t_{hold-MISO}$	200			ns	Data held until next falling SCLK edge
MOSI Hold Time	$t_{hold-MOSI}$	200			ns	Amount of time data is valid after SCLK rising edge
MOSI Setup Time	$t_{setup-MOSI}$	120			ns	From data valid to SCLK rising edge
SPI Time Between Write Commands	t_{SWW}	10.5			μs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second data byte.
SPI Time Between Write And Read Commands	t_{SWR}	6			μs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second address byte.
SPI Time Between Read And Subsequent Commands	t_{SRW} t_{SRR}	1.5			μs	From rising SCLK for last bit of the first data byte, to falling SCLK for the first bit of the address byte of the next command.
SPI Read Address-Data Delay	t_{SRAD}	2			μs	From rising SCLK for last bit of the address byte, to falling SCLK for first bit of data being read.
NCS Inactive After Motion Burst	t_{BEXIT}	500			ns	Minimum NCS inactive time after motion burst before next SPI usage
NCS To SCLK Active	$t_{NCS-SCLK}$	120			ns	From last NCS falling edge to first SCLK rising edge
SCLK To NCS Inactive (For Read Operation)	$t_{SCLK-NCS}$	120			ns	From last SCLK rising edge to NCS rising edge, for valid MISO data transfer
SCLK To NCS Inactive (For Write Operation)	$t_{SCLK-NCS}$	2			μs	From last SCLK rising edge to NCS rising edge, for valid MOSI data transfer
NCS To MISO High-Z	$t_{NCS-MISO}$			500	ns	From NCS rising edge to MISO high-Z state
MOTION Rise Time	$t_{r-MOTION}$		50		ns	$C_L = 100\text{pF}$
MOTION Fall Time	$t_{f-MOTION}$		50		ns	$C_L = 100\text{pF}$
Input Capacitance	C_{in}		50		pF	SCLK, MOSI, NCS
Load Capacitance	C_L			100	pF	MISO, MOTION

Note: All the parameters are tested under operating conditions: VDD = 1.9V, VDDIO = 1.9V, $T_A = 25^\circ\text{C}$.

2.6 Resolution versus Height Chart

This chart serves as a reference of resolution count with its corresponding height based on default setting. The resolution can be changed via register *RESOLUTION* (Section 8.2.2).

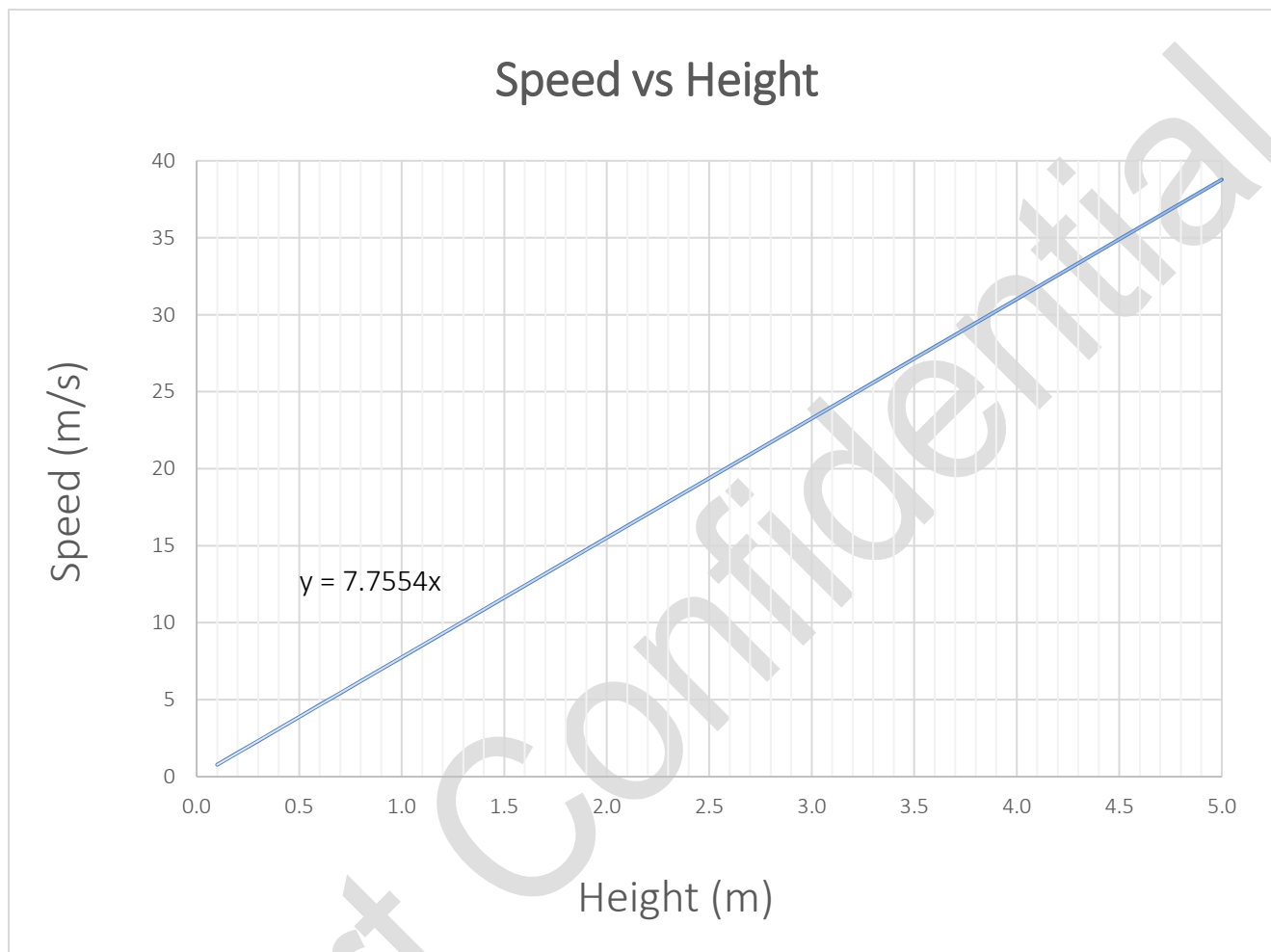


Note: Interpolation is applied to resolution count beyond 2 m.

Figure 3. Resolution versus Height Chart

2.7 Speed versus Height Chart

This chart serves as a theoretical reference of speed capability with its corresponding height for Mode 0 & 1 only.



Note: Interpolation is applied beyond 0.5 m.

Figure 4. Speed versus Height Chart (Mode 0 & 1)

3.0 Mechanical Specifications

3.1 LGA Package Outline Drawing

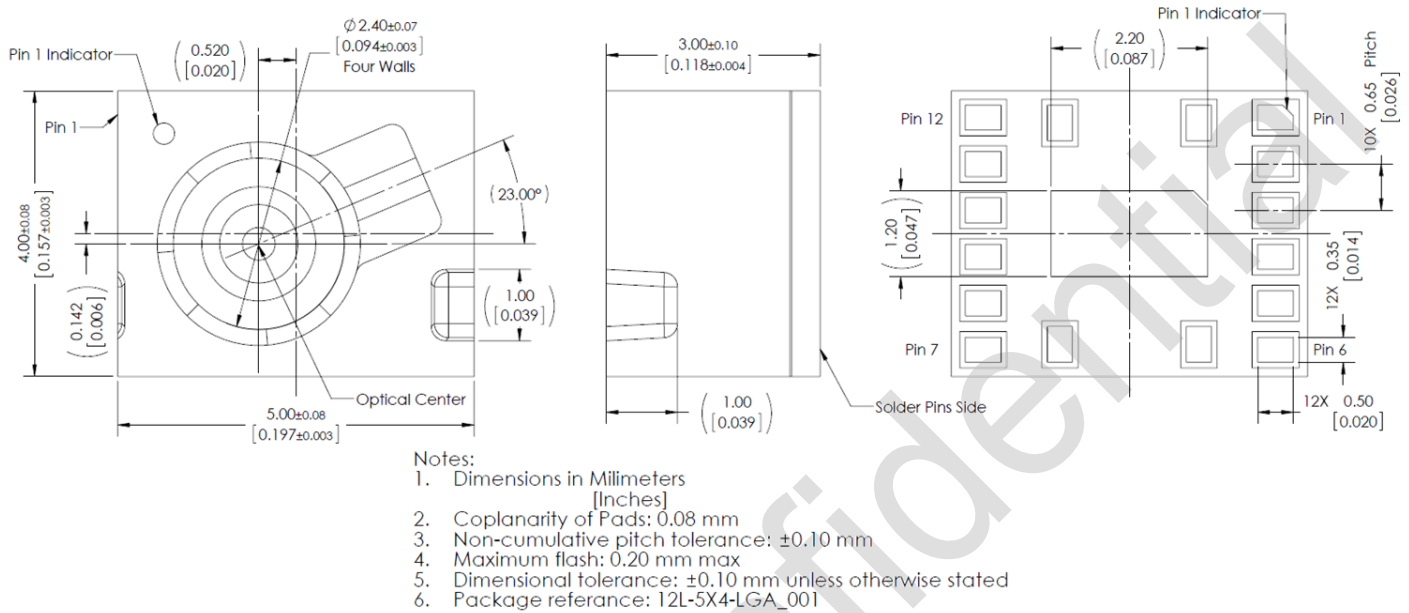


Figure 5. LGA Package Outline Drawing

3.2 Package Marking

Refer Figure 2. Pin Configuration for the code marking location on the device package.

Table 7. Code Identification

Code	Marking	Description
Product Number	PAA3905E1-Q	Part number label
Date Code	YWX	Date label

3.3 Packing Information

Item	Description
Product number	PAA3905E1-Q
Package type	12-pin LGA
Quantity per tube	80 pcs
Inner box quantity	2,000 pcs [25 tubes per inner box]
Shipping box quantity	24,000 pcs [12 inner boxes per shipping box]
Tube size	420 x 5.5 x 4.8 mm ³
Inner box size	89 x 540 x 58 mm ³
Shipping box size	310 x 560 x 270 mm ³

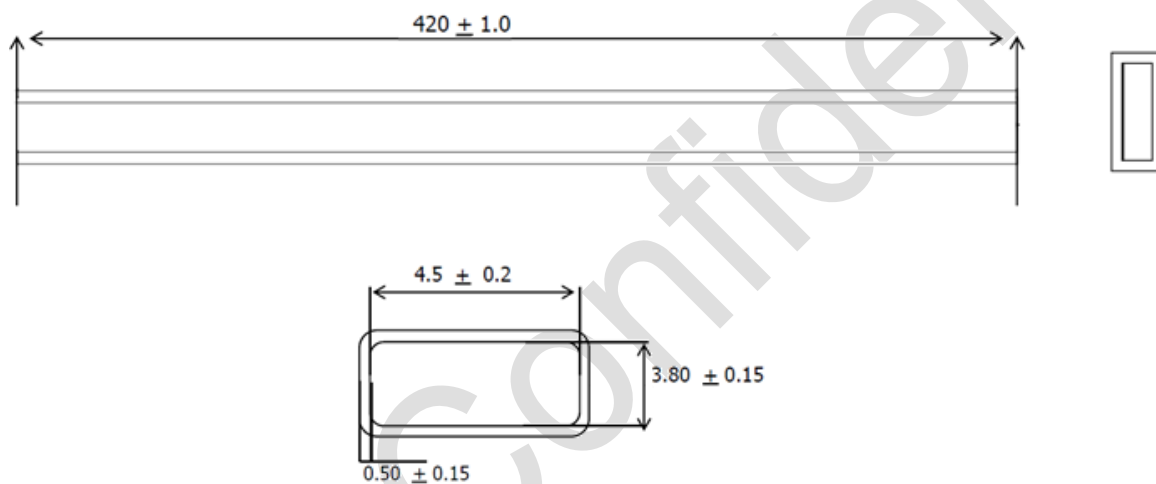


Figure 6. Tube Size (in mm)

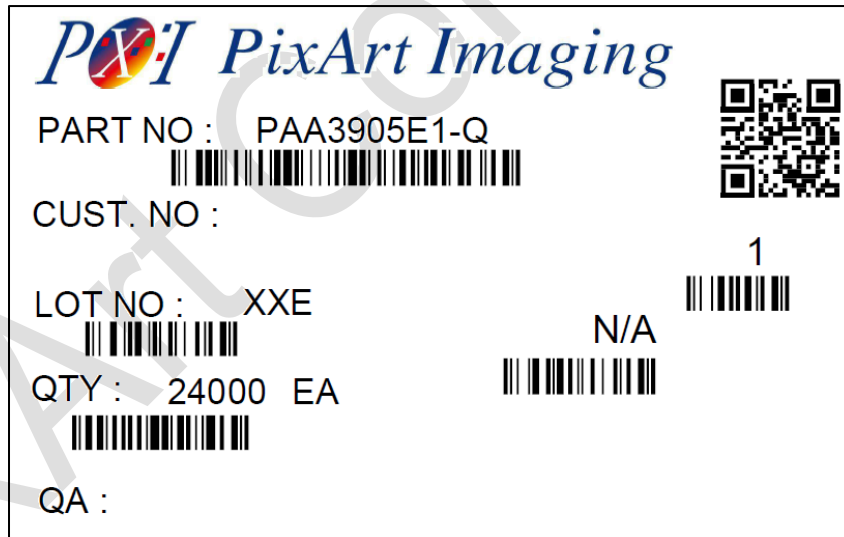
3.4 Package Handling Information

3.4.1 Sample of Inner Box Label



Note: This label is used on inner box

3.4.2 Sample of Shipping Box Label



Note: This label is used on shipping box

4.0 Design References

4.1 General Reference Schematic

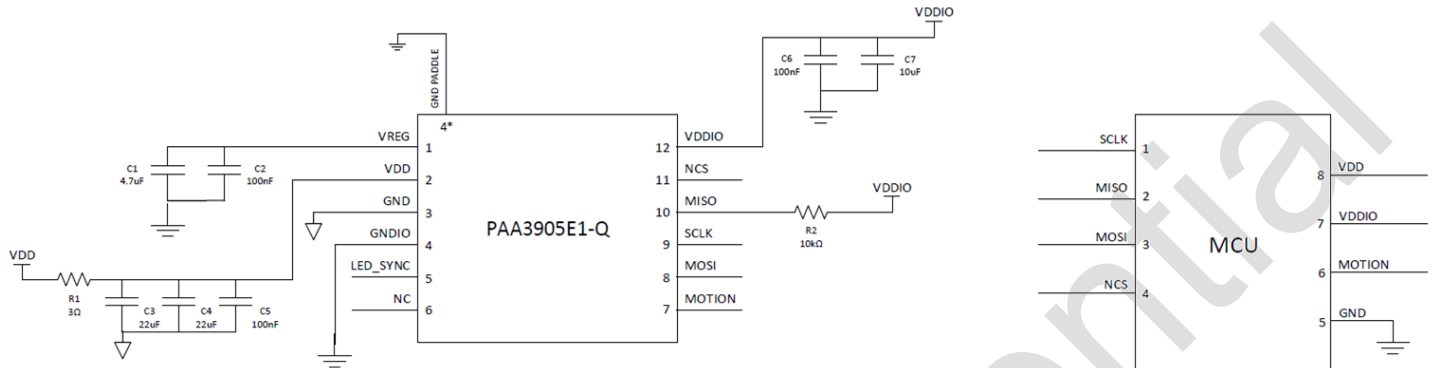


Figure 7. Reference Schematics

4.2 PCB Layout Design Guide

4.2.1 Recommended PCB Footprint

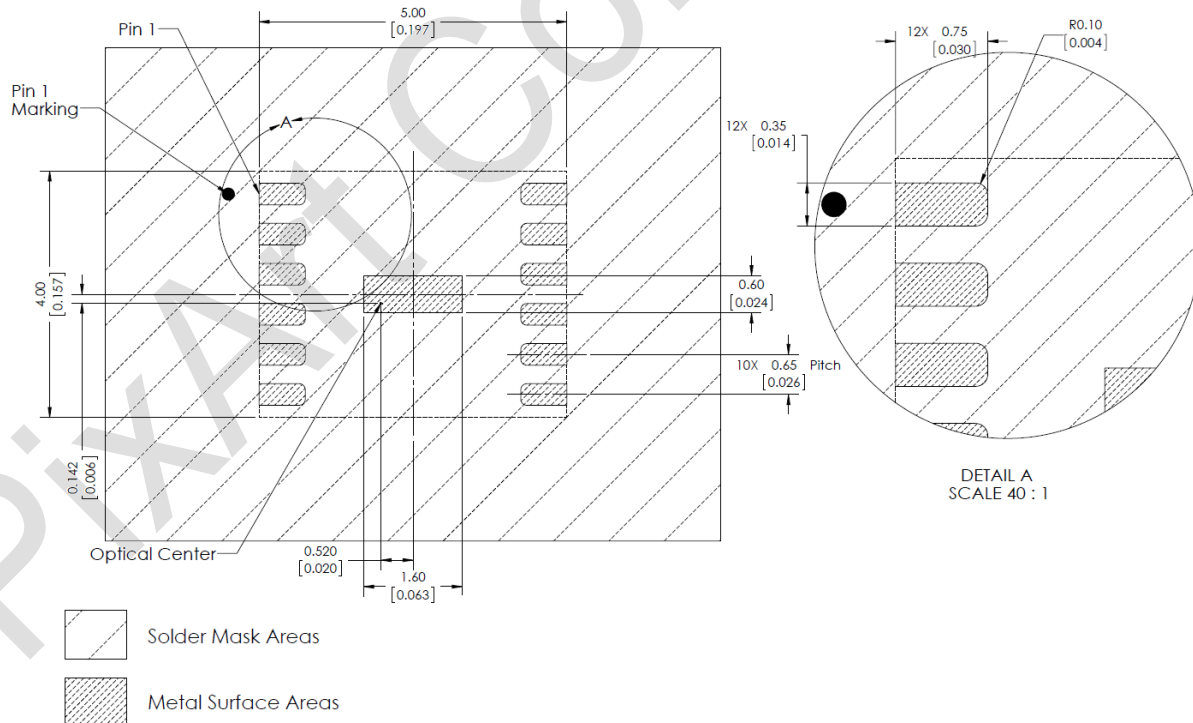


Figure 8. Recommended PCB Footprint

4.3 Assembly Guide

Proper grounding especially handling and assembly of the components in PixArt manufacturing process is required to prevent the chip from damage prematurely or degradation of the chip performance.

4.3.1 IR Reflow Soldering Profile

- Surface mount the chip and all other electrical components onto PCB.
- Reflow the entire assembly in a no-wash solder process.

Note: It is recommended to generate a stencil profile for the reflow process.

- Remove the protective Kapton tape on top of the chip's package.

Note: Avoid contamination towards the cavity surface.

- Recommend to hold the PCB assembly vertically when removing Kapton tape.

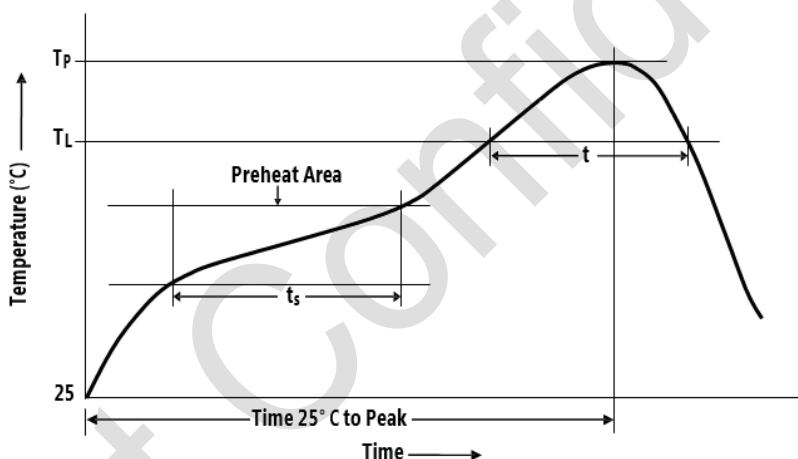


Figure 9. Solder Reflow Profile

Table 8. Reflow Profile

Specification	Value
Max rising slope	0° – 3°C/sec
Preheat duration (150 – 190°C), t_s	60 – 120 sec
Time above reflow ($T_L = 220^\circ\text{C}$), t	30 – 60 sec
Peak temperature, T_p	230 – 260°C

Note: T_L is the Liquids Temperature

4.3.2 Assembly Recommendation

1. Place the PCB assembly horizontally with package cavity facing up. Insert the lens set with plastic or soft-tip tweezer onto the optical aperture (the chip's cavity).
2. Insert the lens set onto the chip package cavity (the hole on the chip's package).
3. Use an appropriate flat tip jig to press the lens barrel vertically onto the upper cavity of the chip's package.

Note: No lens calibration is required.

4. Insert the nozzle of glue dispenser vertically inside the gluing slots and dispense glue appropriately.
5. Remove the nozzle of glue dispenser and let the glue cure properly.

Note: Refer L242-ZSZ1 lens set datasheet for more information and detailed steps of the assembly process.

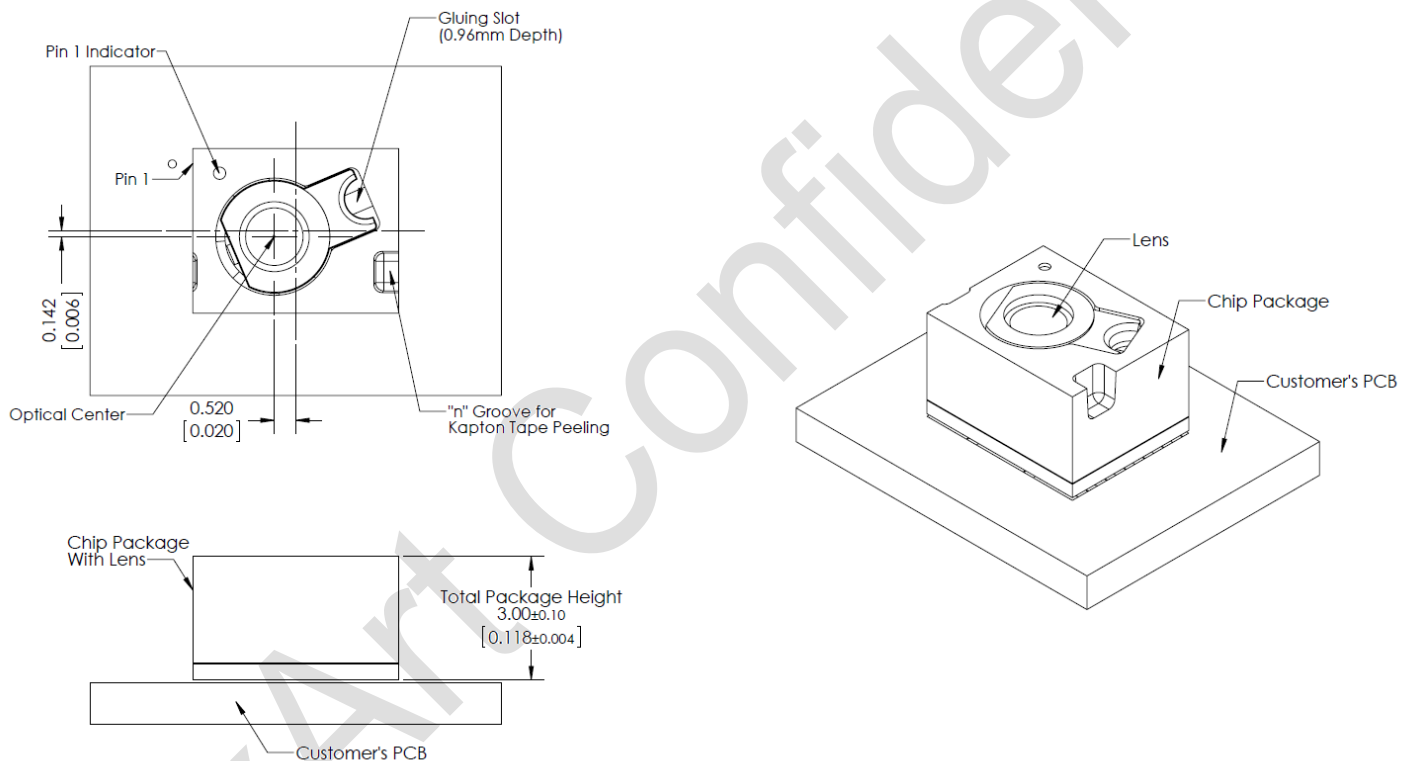


Figure 10. System Assembly View with L242-ZSZ1

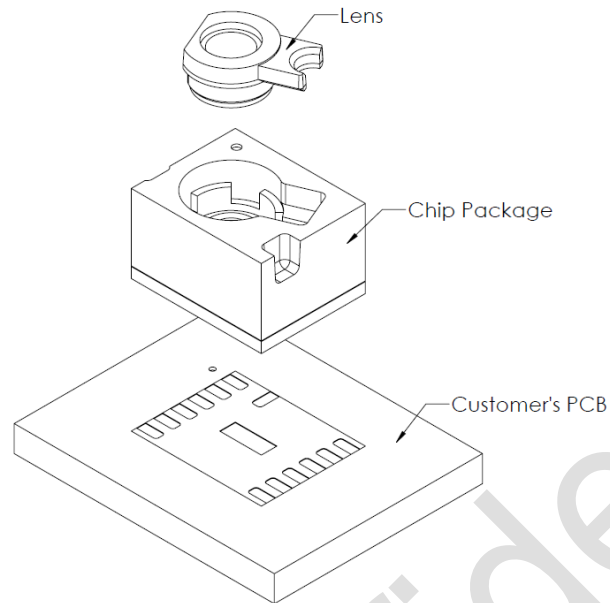


Figure 11. Exploded View of System Assembly

4.3.3 Manual rework of chip assembly

This is an optional step.

For de-solder chip package from PCB, do disassemble the lens set first as it melts under the soldering heat.

Please note below precautions for re-work of chip assembly.

- Pry up lens set from the chip package cavity from the gluing slot using a tweezer.

Note: It is important to remove the lens set as it may melt under the soldering heat.

- Place kapton tape across the top of the chip package cavity to avoid contamination
- Perform de-soldering & soldering activities as needed.
- Remove kapton tape and insert the lens set as outlined in the above section.

5.0 Power Management

5.1 Power Supply

The chip has two power supply inputs (VDD and VDDIO). VDD is the main power supply and VDDIO is the I/O reference voltage.

5.2 Power Sequence

5.2.1 Power on Sequence Requirement

VDDIO need to be powered up earlier than VDD, the interval time between VDD and VDDIO must be less than 100ms

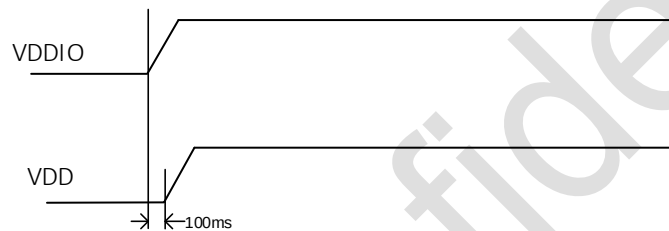


Figure 12. Power-up Sequence Requirement

5.2.2 Power off Sequence Requirement

It is recommended to power off both VDD and VDDIO at the same time or the VDD first, then follow by VDDIO.

5.3 Power States

5.3.1 States Description

Table 9. States Description

State	Functional Description
OFF	No power supply, all the voltage rails and clocks are gated.
RUN	The chip enters RUN State after the regular power on and initialization.
Shutdown	The chip enters this state upon receive host command for power saving purpose and all the parameter settings still intact. The chip exits shutdown state and transition back to RUN state upon receive host command. To exit Shutdown state, only power up command (write <i>Shutdown</i> register with value 0xC7) is accepted.

The table below shows the state of various pins during shutdown.

Table 10. State of Signal Pins during Shutdown.

Pin	Status during Shutdown State
NCS	High ¹
MISO	Hi-Z ²
SCLK	Ignore if NCS = 1 ³
MOSI	Ignore if NCS = 1 ⁴
MOTION	Output High

5.3.2 State Diagram

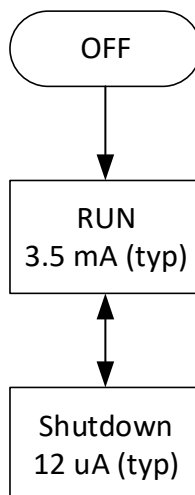


Figure 13. Power States Diagram

5.3.3 State Transition

State	Type	Description
OFF to RUN	Power-up	After the chip is powered on and initialization is completed, the chip enters RUN state automatically.
RUN to Shutdown	Command	The host to send command to enter Shutdown state
Shutdown to RUN	Command	The host to send command to exit Shutdown state

▪ RUN to Shutdown

Write register 0x3B with value 0xB6

// Enter Shutdown state

▪ Shutdown to RUN

Step in sequence as follow:

1. Drive NCS high for 1ms, and then low // To reset the SPI port
2. Write register 0x3B with value 0xC7 // Exit Shutdown state command
3. Wait for at least 1ms.
4. Write register 0x3B with value 0x00
5. Wait for at least 1ms.
6. Read from registers 0x02, 0x03, 0x04, 0x05 and 0x06 // read once to clear the registers

5.4 Reset

The table below shows the state of the various pins during power-up and reset.

Table 11. State of Signal Pins during Power-Up & Reset

State of Signal Pins after V_{DD} is Valid		
Pin	During Reset	After Reset
NCS	Ignored	Functional
MISO	Undefined	Depends on NCS
SCLK	Ignored	Depends on NCS
MOSI	Ignored	Depends on NCS
MOTION	Undefined	Functional

5.4.1 Power on Reset

During power-on, the chip does not need an external reset as there is an internal circuitry that performs power-on reset function for the chip.

5.4.2 Software Reset

Write 0x5A to the *Power_Up_Reset* register to reset the chip. All register settings will be reverted to default values, the chip is required for re-initialization.

5.5 Related Registers

Usage	Name	Bank	Address
Power Management	<i>Shutdown Register</i>	-	0x3B
	<i>Power_Up_Reset</i>	-	0x3A

■ Shutdown Register

Register Name	Shutdown		
Bank	-	Address	0x3B
Access	W	Default Value	-
Description	Shutdown state control register. Refer Section 5.3.3 for the details.		

■ Reset Register

Register Name	Power_Up_Reset		
Bank	-	Address	0x3A
Access	W	Default Value	-
Description	Write 0x5A to this register to reset the chip. All settings will revert to default values. Reset is required to restore normal operation after Raw Data Output.		

6.0 Control Interface

The chip adopts Serial Port Interface Communication (SPI) interface. The interface is configured to mode 3 (CPHA=1 and CPOL=1) for slave device communication. The maximum SCLK speed is 2MHz.

6.1 Signal Description

The 4-wire synchronous serial port interface is used to write or read registers in the chip. The host is a SPI master device where it drives SCLK, MOSI, and NCS (active low) to initiate the communication. The interface supports single or multi-chip (slaves) with NCS control.

Table 12. Signal Description

Pin	Reset Status	Description
SCLK	High	Clock input, generated by the host (master).
MOSI	High	Input data (Master Out / Slave In).
MISO	Hi-Z	Output data (Master In / Slave Out).
NCS	High	Chip select input (active low).

6.2 Chip Select Operation

The serial port is activated after NCS is asserted. If NCS is de-asserted during a transaction, the entire transaction is aborted and the serial port will be reset. After a transaction is aborted, the host needs to follow the normal address-to-data or transaction-to-transaction delay timing requirement before beginning the next transaction.

When the NCS pin is high, the SPI's inputs are ignored and the SPI's output are in tri-state. NCS can also be used to reset the serial port communication in case of an error occur.

To improve communication reliability, all serial transactions should be framed by NCS assertion. The port should not be remained enabled when not in use to prevent misinterpret ESD and EFT/B events and put the chip into an unknown state.

6.3 Protocol

The transmission protocol is a 4-wire link, half duplex protocol between the host and the chip. All data changes at SCLK falling edge and latched at SCLK rising edge. The host controller always initiates communication and the chip never initiates data transfers.

The transmission protocol consists of the two type operations:

- Write Operation
- Read Operation

Both of the two operation modes consist of two bytes. The first byte contains the register address (seven bits) and bit 7 as MSB to indicate data direction. The second byte contains the data.

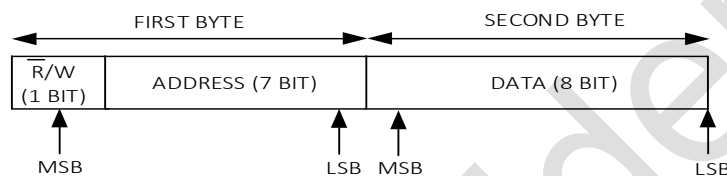


Figure 5. Transmission Protocol

6.4 Write Operation

The write operation consists of two bytes package. The first byte contains the 7-bit register address and a “1” in bit [7] indicates data write direction from the host to the chip. The second byte contains the data to be written into the specified register. The chip latches MOSI data on rising edges of the SCLK.

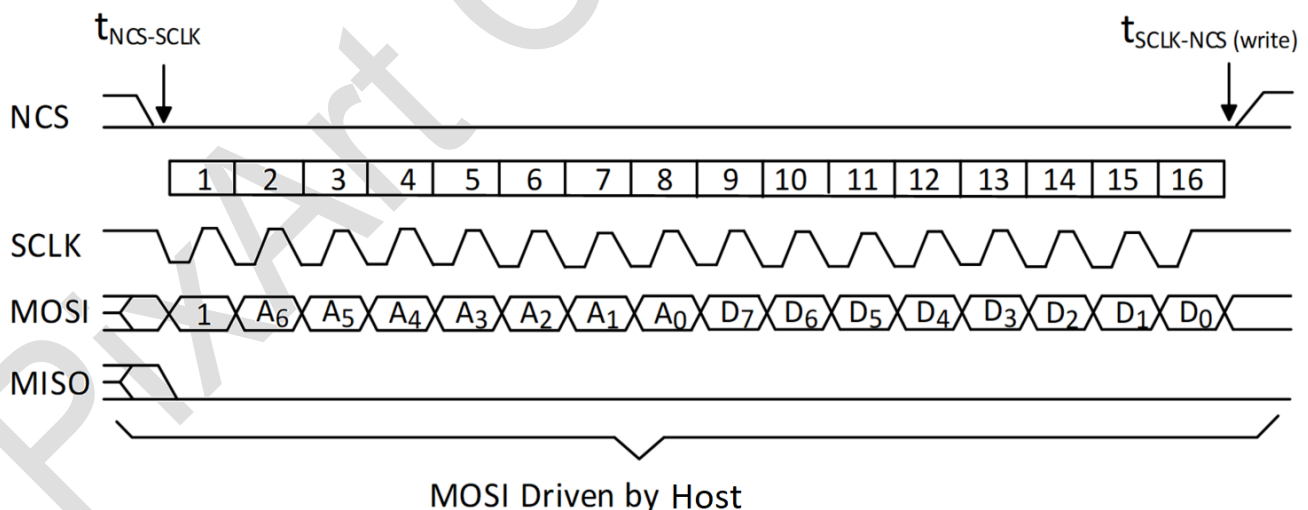


Figure 14. Write Operation

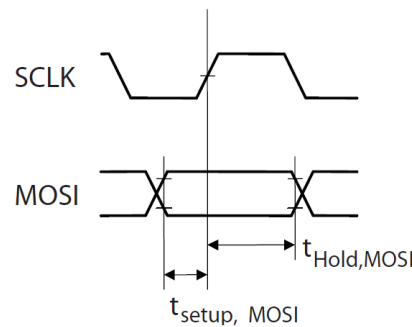


Figure 15. MOSI Set-up and Hold Time

6.5 Read Operation

The read operation consists of two bytes package. The first byte contains the 7-bit register address and a “0” in bit [7] indicates data read direction from the chip to the host and the chip samples MOSI bits on every rising edge of SCLK. The second byte contains the data from the specified register and the chip outputs MISO bits on falling edges of SCLK.

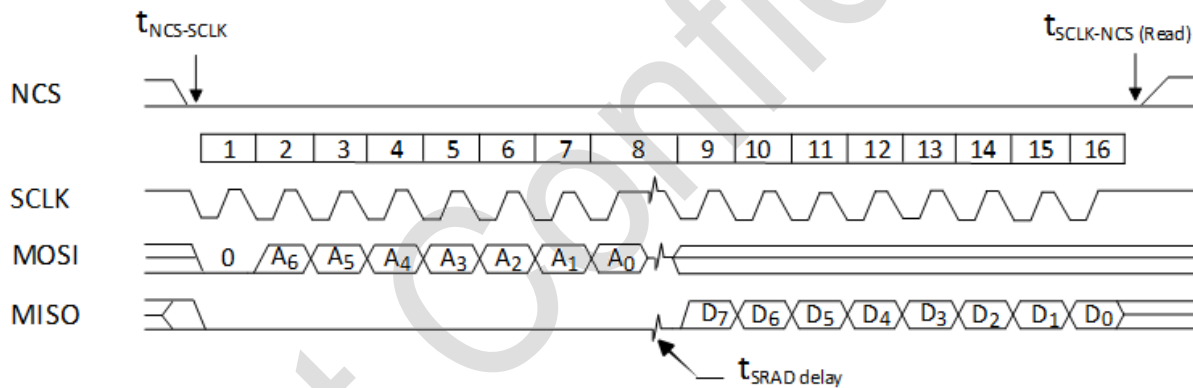


Figure 16. Read Operation

The minimum high state of SCLK is also the minimum MISO data hold time of the chip. Since the falling edge of SCLK is actually the start of the next read or write command, the chip will hold the state of data on MISO until the falling edge of SCLK.

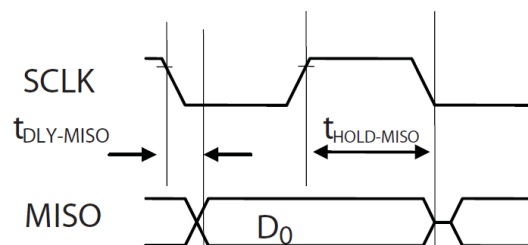


Figure 17. MISO Delay and Hold Time

6.6 Burst Read

Burst read is a special serial port interface operation mode which can be used to reduce the serial transaction time for Motion Read. The speed improvement is achieved by continuous data clocking to read multiple registers with a single dedicated address and not requiring the delay period between data bytes. Refer Section 7.3.1 for the detail.

NCS must be de-asserted to terminate burst mode after each burst read operation transaction is completed. However, the host can terminate the Burst Read operation as needed through de-assert the NCS.

After sending the register address, the host must wait for t_{SRAD} , and then begin reading data. All data bits can be read with no delay between bytes by driving SCLK at the normal rate. The data are latched into the output buffer after the last address bit is received. After the burst transmission is completed, the host NCS is de-assert for at least t_{BEXIT} to terminate burst read operation. The serial port is not available for use until assert of NCS.

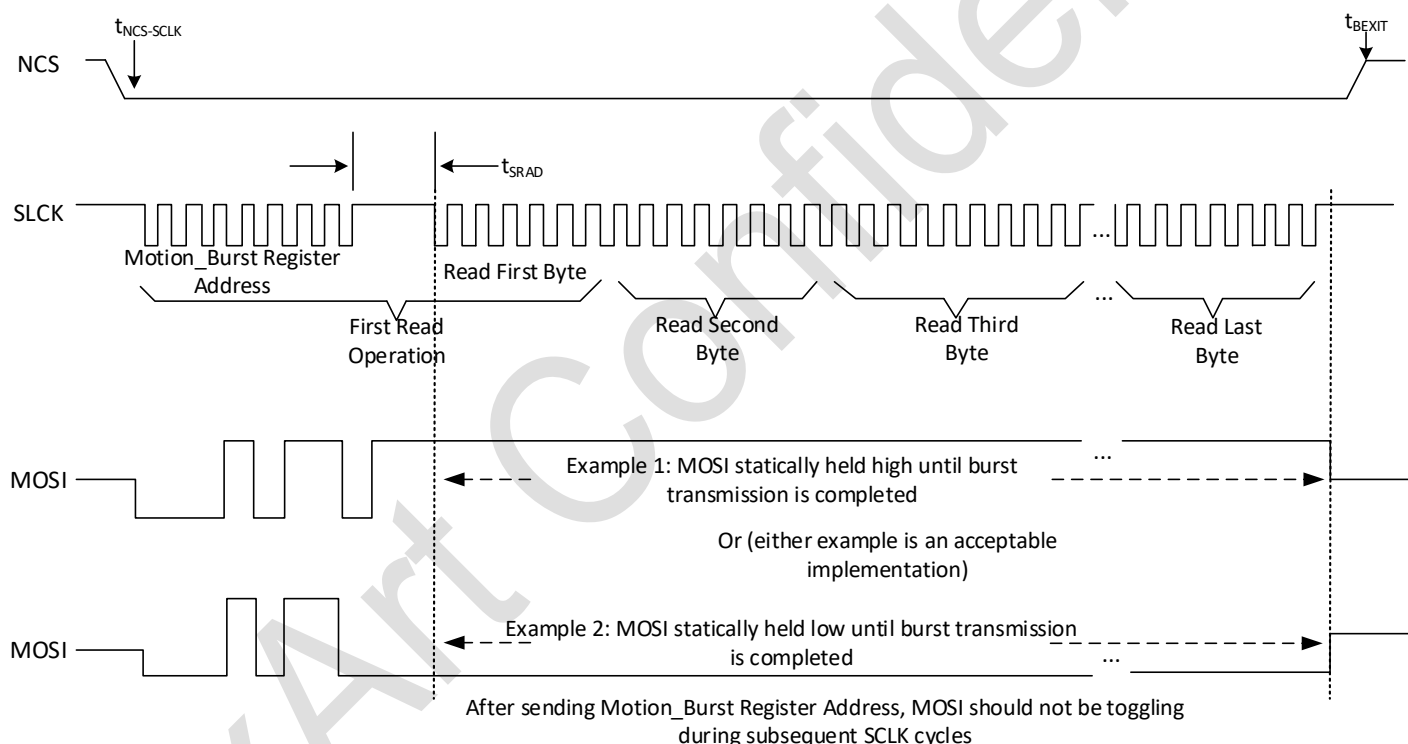


Figure 18. Burst Read Timing

6.7 Required Timing between Read and Write Commands (t_{sxx})

There are minimum timing requirements between read and write commands on the serial port.

If the rising edge of the SCLK for the last data bit of the second write command occurs before the t_{sww} delay, then the first write command may not complete correctly.

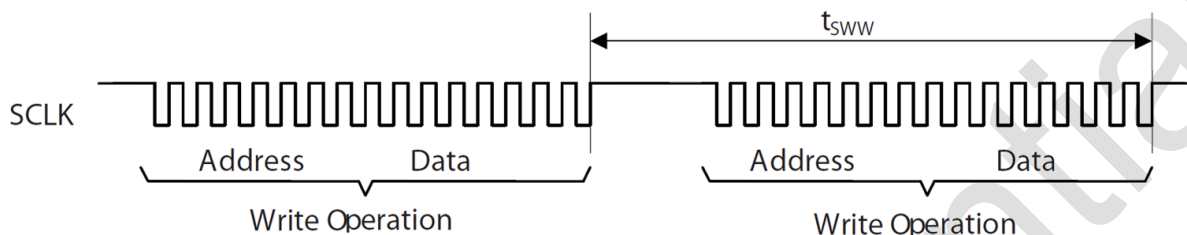


Figure 19. Timing between two Write Commands

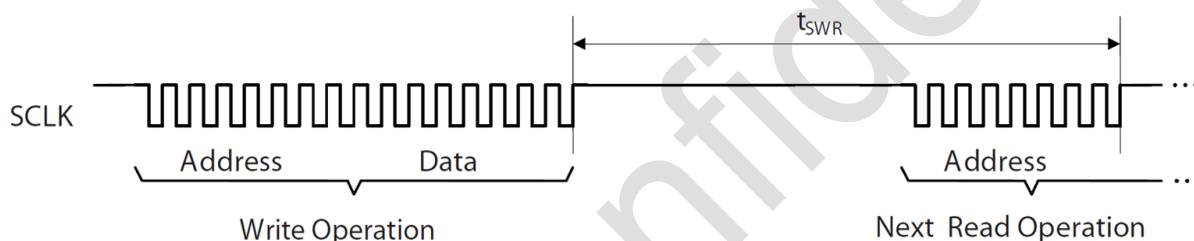


Figure 20. Timing between Write and Read commands

If the rising edge of SCLK for the last address bit of the read command occurs before the t_{swr} required delay, the write command may not complete correctly. During a read operation, SCLK should be delayed at least t_{srad} after the last address data bit to ensure that the chip has time to prepare the requested data.

The falling edge of SCLK for the first address bit of either the read or write command must be at least t_{srr} or t_{srw} after the last SCLK rising edge of the last data bit of the previous read operation. In addition, during a read operation SCLK should be delayed after the last address data bit to ensure that the chip has time to prepare the requested data.

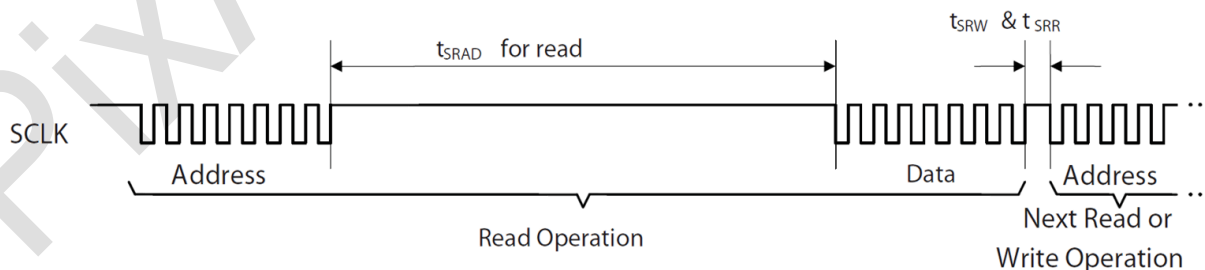


Figure 21. Timing between Read and either Write or subsequent Read commands

7.0 System Control

7.1 System Initialization

7.1.1 Initial Flow

Although the chip performs an internal self-power on reset, it is still recommended that the *Power_Up_Reset* register is written every time power is applied. The appropriate sequence is as follows:

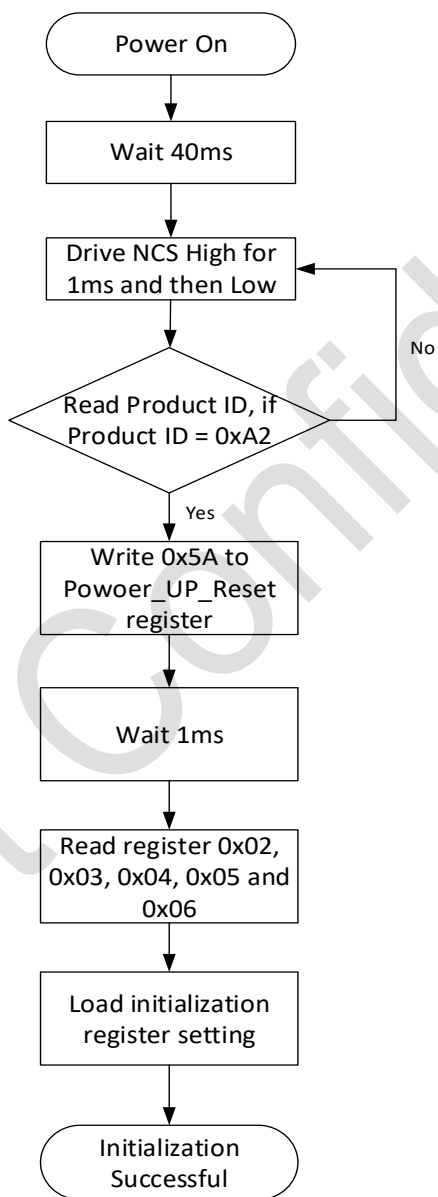


Figure 22. System Initialization Flow Chart

Note: If unable to read the correct Product ID, it is recommended to power off and power on the Module again. If the problem still persist, please check the SPI interface connection and settings.

1. Apply power to V_{DDIO} first and followed by V_{DD} , with a delay of no more than 100ms in between each supply. Ensure all supplies are stable.
2. Wait for at least 40 ms.
3. Drive NCS high for 1ms, and then low to reset the SPI port.
4. Read the *Product_ID* register to verify the PID before any other register read or write.
5. Write 0x5A to *Power_Up_Reset* register.
6. Wait for at least 1 ms.
7. Read from registers 0x02, 0x03, 0x04, 0x05 and 0x06 one time regardless of the motion pin state to clear the motion data.
8. Refer Section 7.1.2 to configure the needed registers in order to achieve optimum performance of the chip.

7.1.2 Initialization Register Setting

Upon power-up of the chip, there are a number of registers to be configured to initialize the chip. There are two types of Detection Mode Setting, Standard Detection Mode Setting and Enhanced Detection Mode Setting.

- Standard Detection Setting is recommended for general tracking operations. In this mode, the chip can detect when it is operating over striped, checkerboard, and glossy tile surfaces where tracking performance is compromised.
- Enhance Detection Setting relatively has better detection sensitivity, it is recommended where yaw motion detection is required, and also where more sensitive challenging surface detection is required. The recommended operating height must be greater than 15 cm to avoid false detection on challenging surfaces due to increasing of sensitivity.

■ Standard Detection Setting

Recommend to perform software reset before loading below mode switching.

1. Write register 0x7F with value 0x00
2. Write register 0x51 with value 0xFF
3. Write register 0x4E with value 0x2A
4. Write register 0x66 with value 0x3E
5. Write register 0x7F with value 0x14
6. Write register 0x7E with value 0x71
7. Write register 0x55 with value 0x00
8. Write register 0x59 with value 0x00
9. Write register 0x6F with value 0x2C
10. Write register 0x7F with value 0x05
11. Write register 0x4D with value 0xAC
12. Write register 0x4E with value 0x32
13. Write register 0x7F with value 0x09
14. Write register 0x5C with value 0xAF
15. Write register 0x5F with value 0xAF
16. Write register 0x70 with value 0x08
17. Write register 0x71 with value 0x04
18. Write register 0x72 with value 0x06
19. Write register 0x74 with value 0x3C
20. Write register 0x75 with value 0x28
21. Write register 0x76 with value 0x20
22. Write register 0x4E with value 0xBF
23. Write register 0x7F with value 0x03
24. Write register 0x64 with value 0x14
25. Write register 0x65 with value 0x0A
26. Write register 0x66 with value 0x10
27. Write register 0x55 with value 0x3C
28. Write register 0x56 with value 0x28
29. Write register 0x57 with value 0x20
30. Write register 0x4A with value 0x2D
31. Write register 0x4B with value 0x2D
32. Write register 0x4E with value 0x4B
33. Write register 0x69 with value 0xFA
34. Write register 0x7F with value 0x05
35. Write register 0x69 with value 0x1F
36. Write register 0x47 with value 0x1F
37. Write register 0x48 with value 0x0C
38. Write register 0x5A with value 0x20
39. Write register 0x75 with value 0x0F
40. Write register 0x4A with value 0x0F
41. Write register 0x42 with value 0x02
42. Write register 0x45 with value 0x03
43. Write register 0x65 with value 0x00
44. Write register 0x67 with value 0x76
45. Write register 0x68 with value 0x76
46. Write register 0x6A with value 0xC5
47. Write register 0x43 with value 0x00
48. Write register 0x7F with value 0x06
49. Write register 0x4A with value 0x18
50. Write register 0x4B with value 0x0C
51. Write register 0x4C with value 0x0C
52. Write register 0x4D with value 0x0C
53. Write register 0x46 with value 0x0A
54. Write register 0x59 with value 0xCD
55. Write register 0x7F with value 0x0A
56. Write register 0x4A with value 0x2A
57. Write register 0x48 with value 0x96
58. Write register 0x52 with value 0xB4
59. Write register 0x7F with value 0x00
60. Write register 0x5B with value 0xA0

▪ Enhanced Detection Mode

Recommend to perform software reset before loading below settings.

1. Write register 0x7F with value 0x00
2. Write register 0x51 with value 0xFF
3. Write register 0x4E with value 0x2A
4. Write register 0x66 with value 0x26
5. Write register 0x7F with value 0x14
6. Write register 0x7E with value 0x71
7. Write register 0x55 with value 0x00
8. Write register 0x59 with value 0x00
9. Write register 0x6F with value 0x2C
10. Write register 0x7F with value 0x05
11. Write register 0x4D with value 0xAC
12. Write register 0x4E with value 0x65
13. Write register 0x7F with value 0x09
14. Write register 0x5C with value 0xAF
15. Write register 0x5F with value 0xAF
16. Write register 0x70 with value 0x00
17. Write register 0x71 with value 0x00
18. Write register 0x72 with value 0x00
19. Write register 0x74 with value 0x14
20. Write register 0x75 with value 0x14
21. Write register 0x76 with value 0x06
22. Write register 0x4E with value 0x8F
23. Write register 0x7F with value 0x03
24. Write register 0x64 with value 0x00
25. Write register 0x65 with value 0x00
26. Write register 0x66 with value 0x00
27. Write register 0x55 with value 0x14
28. Write register 0x56 with value 0x14
29. Write register 0x57 with value 0x06
30. Write register 0x4A with value 0x20
31. Write register 0x4B with value 0x20
32. Write register 0x4E with value 0x32
33. Write register 0x69 with value 0xFE
34. Write register 0x7F with value 0x05
35. Write register 0x69 with value 0x14
36. Write register 0x47 with value 0x14
37. Write register 0x48 with value 0x1C
38. Write register 0x5A with value 0x20
39. Write register 0x75 with value 0xE5
40. Write register 0x4A with value 0x05
41. Write register 0x42 with value 0x04
42. Write register 0x45 with value 0x03
43. Write register 0x65 with value 0x00
44. Write register 0x67 with value 0x50
45. Write register 0x68 with value 0x50
46. Write register 0x6A with value 0xC5
47. Write register 0x43 with value 0x00
48. Write register 0x7F with value 0x06
49. Write register 0x4A with value 0x1E
50. Write register 0x4B with value 0x1E
51. Write register 0x4C with value 0x34
52. Write register 0x4D with value 0x34
53. Write register 0x46 with value 0x32
54. Write register 0x59 with value 0x0D
55. Write register 0x7F with value 0x0A
56. Write register 0x4A with value 0x2A
57. Write register 0x48 with value 0x96
58. Write register 0x52 with value 0xB4
59. Write register 0x7F with value 0x00
60. Write register 0x5B with value 0xA0

7.2 Operation Modes

Upon the chip power on as per the recommended power on sequence in Section 7.1.1, the chip automatically switches among three operation modes when operating in different brightness conditions. The three operation modes are to cater for different far field application needs and ambient conditions. The modes are described below.

Table 13. Three Operation Modes

Mode	Frame Rate (Typ)	Description	Lux (Typ)
0	126 fps	Bright Mode for general motion tracking	60
1	126 fps	Low Light Mode for low light motion tracking	30
2	50 fps	Super Low Light Mode for super low light and low speed motion tracking	5

7.2.1 Automatic Switching of Operation Modes

The chip automatically switches in between three operation modes when operating under different brightness conditions. To enable the automatic switching of operation modes, perform below register writes in sequence. Please note that the chip enables the automatic switching between Mode 0, 1 and 2 by default after loading the initialization register setting. In the event of the sensor is switched to manual switching mode (Section 7.2.2), the automatic switching between modes can be re-enable by performing below register writes.

- Automatic switching between Mode 0, 1 and 2:
 1. Write register 0x7F with value 0x08
 2. Write register 0x68 with value 0x02
 3. Write register 0x7F with value 0x00
- Automatic switching between Mode 0 and 1 only:
 1. Write register 0x7F with value 0x08
 2. Write register 0x68 with value 0x01
 3. Write register 0x7F with value 0x00

7.2.2 Manual Switching of Operation Modes

The operation mode can be manually switched depending on user's specific application. Perform below register writes to set to the desired operation mode.

- Mode 0

1. Write register 0x7F with value 0x08
2. Write register 0x68 with value 0x00
3. Write register 0x7F with value 0x00

- Mode 1

1. Write register 0x7F with value 0x08
2. Write register 0x68 with value 0x11
3. Write register 0x7F with value 0x00

- Mode 2

1. Write register 0x7F with value 0x08
2. Write register 0x68 with value 0x22
3. Write register 0x7F with value 0x00

Note: While this section aims to provide flexibility for the user to select the operation mode. User owns the responsibility to select the appropriate operation mode to meet the desired end application.

7.3 Registers Access

The host can access register through SPI interface. Refer to section 6.0 for detail. The register address is a 7 bits width address mapping (0x00 to 0x7F).

7.3.1 Burst Read Operation

Burst read operation is a special serial port interface operation mode which use to reduce the serial transaction time for motion read. The speed improvement is achieved by continuous data clocking from the host for multiple registers read without further specify the register address. In the burst read operation, NCS must be remained asserted until burst mode transaction is fully completed.

Note: A single read of any motion related registers (0x02 to 0x06) should be avoided during burst mode.

Read the *Motion_Burst* register activates burst mode. The chip will respond with the following motion burst report in order.

Motion burst report:

BYTE[00] = Motion
BYTE[01] = Observation
BYTE[02] = Delta_X_L
BYTE[03] = Delta_X_H
BYTE[04] = Delta_Y_L
BYTE[05] = Delta_Y_H
BYTE[06] = Reserved
BYTE[07] = SQUAL
BYTE[08] = RawData_Sum
BYTE[09] = Maximum_RawData
BYTE[10] = Minimum_RawData
BYTE[11] = Shutter_Upper
BYTE[12] = Shutter_Middle
BYTE[13] = Shutter_Lower

Procedure to start motion burst:

1. Lower NCS signal, and wait for $t_{NCS-SCLK}$ delay.
2. Send *Motion_Burst* address (0x16). After sending this address, MOSI should be held either high or low until the burst transmission is complete (MOSI should not be toggling during subsequent SCLK cycles).
3. Wait for t_{SRAD} .
4. Start reading SPI Data continuously up to 14 bytes.

5. The NCS is de-assert after the burst transmission is complete. Alternatively, motion burst can be terminated by pulling NCS high as needed. However, the NCS de-assert interval need to be at least t_{BEXIT} before next operation.
6. Access SQUAL & Shutter values to check for validity of the motion data. Refer Section 7.4.3 for detail.
7. To read new motion burst data, repeat from step 1.

7.3.2 Related Registers

Usage	Name	Bank	Address
Burst Read	<i>Motion_Burst</i>	-	0x10

Register Name	Motion_Burst		
Bank		Address	0x16
Access	R	Default Value	0x00
Description	The <i>Motion_Burst</i> register is used for high-speed access of up to 12 register bytes. Refer Section 7.3.1 for use details.		

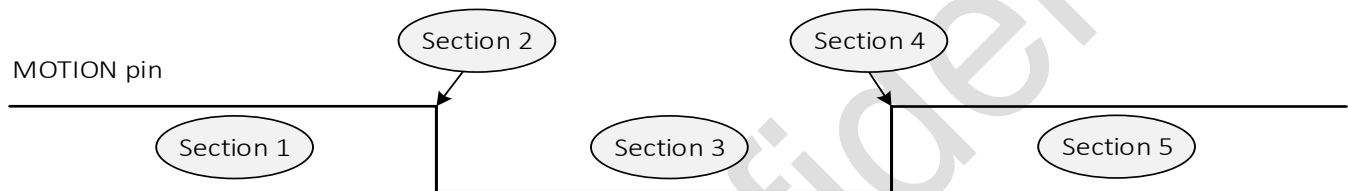
7.4 Output

7.4.1 Motion Bit and Motion Pin Interrupt

The motion bit in the *Motion* register (address 0x02) is an indicator flag for motion detection. Prior reading motion data registers, the host must check motion bit in the *Motion* register (address 0x02). Motion bit will be updated as 1 if the motion data in the motion registers are available.

The motion pin is an interrupt (active low) pin that triggers the host when motion has occurred. The MOTION pin goes low whenever the motion bit in *Motion* register (address 0x02) is "1".

Note: No pull-up resistor is required at the MOTION pin or at the host side to reduce power when motion pin is in low state.



Section 1 : No motion detected, motion data are invalid. Motion bit=0 and motion pin de-asserted.

Section 2 : When motion detected, motion pin is asserted, motion bit=1 and motion data accumulation begins.

Section 3 : Motion data accumulation in progress and wait for host to access.

Section 4 : Upon motion data register is accessed, motion bit=0, interrupt pin is de-asserted and motion data registers are automatic cleared.

Section 5 : No motion detected, motion data are invalid. Motion bit=0 and motion pin de-asserted.

Figure 23. Motion Interrupt Pin Function

7.4.2 Output Access

The accumulated motion data are stored in *Delta_X_L*, *Delta_X_H*, *Delta_Y_L* and *Delta_Y_H* registers. The data can be accessed through single or burst read operation as needed.

Single read access the motion data by using normal read operation when the data in register *Delta_X_L*, *Delta_X_H*, *Delta_Y_L* and *Delta_Y_H* are available. Burst mode operation is a special serial port operation mode which use to reduce the serial transaction time for motion read, refer Section 7.3.1 for the detail.

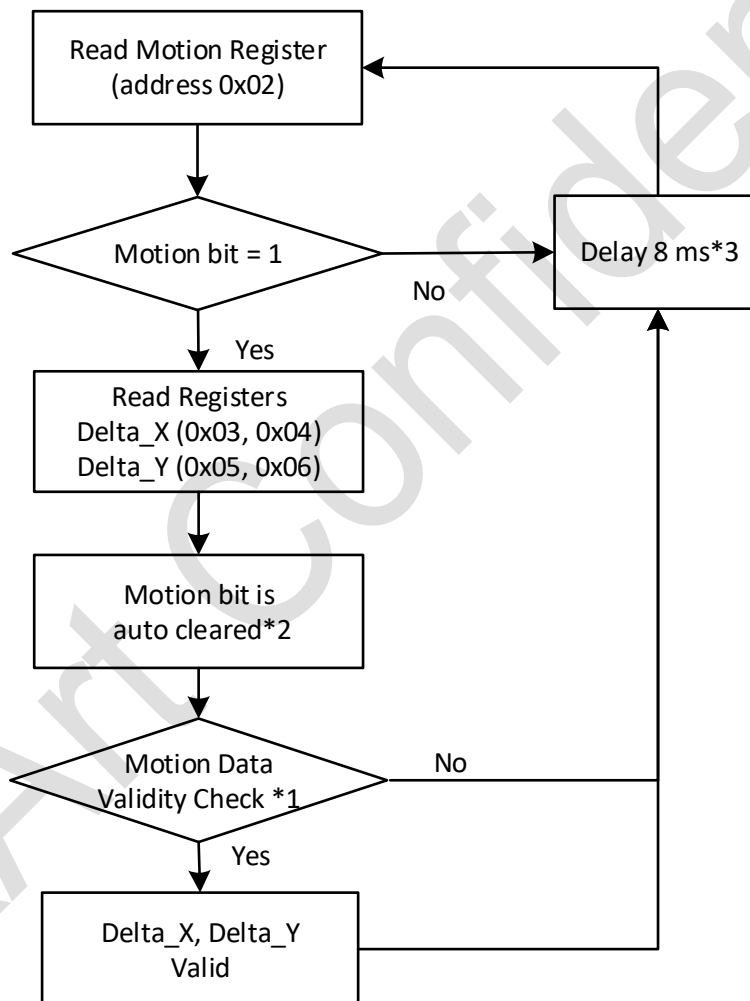
There is two methods to read out the motion data:

- Polling method
- Interrupt method

■ Polling Method

The motion bit in *Motion* register (address 0x02) is an output that signals the host when motion has occurred. The motion bit turns to “1” whenever the motion is detected; in other words, whenever there is non-zero data in the *Delta_X_L*, *Delta_X_H*, *Delta_Y_L* or *Delta_Y_H* registers. Motion bit is cleared to “0” by reading or writing to *MOTION* register.

By reading and checking *Motion* register periodically, the host can get the motion data simply through SPI interface. This register allows user to determine if motion has occurred since the last time it was read. If the Motion register bit-7 = 1, the host can read register 0x03, 0x04, 0x05 and 0x06 to get the accumulated motion data.



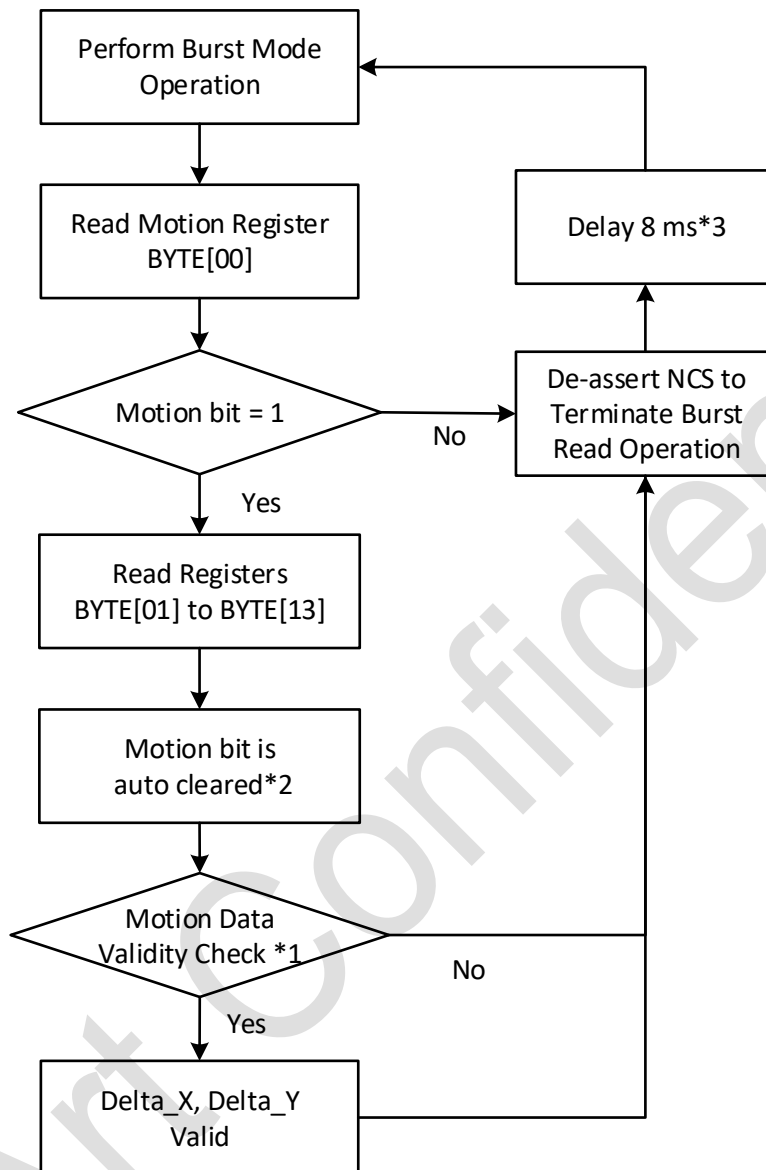
Note:

*1: Refer Section 7.4.3 for detail

*2: Motion bit is auto cleared after reading *MOTION* register

*3: The polling interval is configured by the host

Figure 24. Polling Method – Single Read



Note:

*1: Refer Section 7.4.3 for the detail

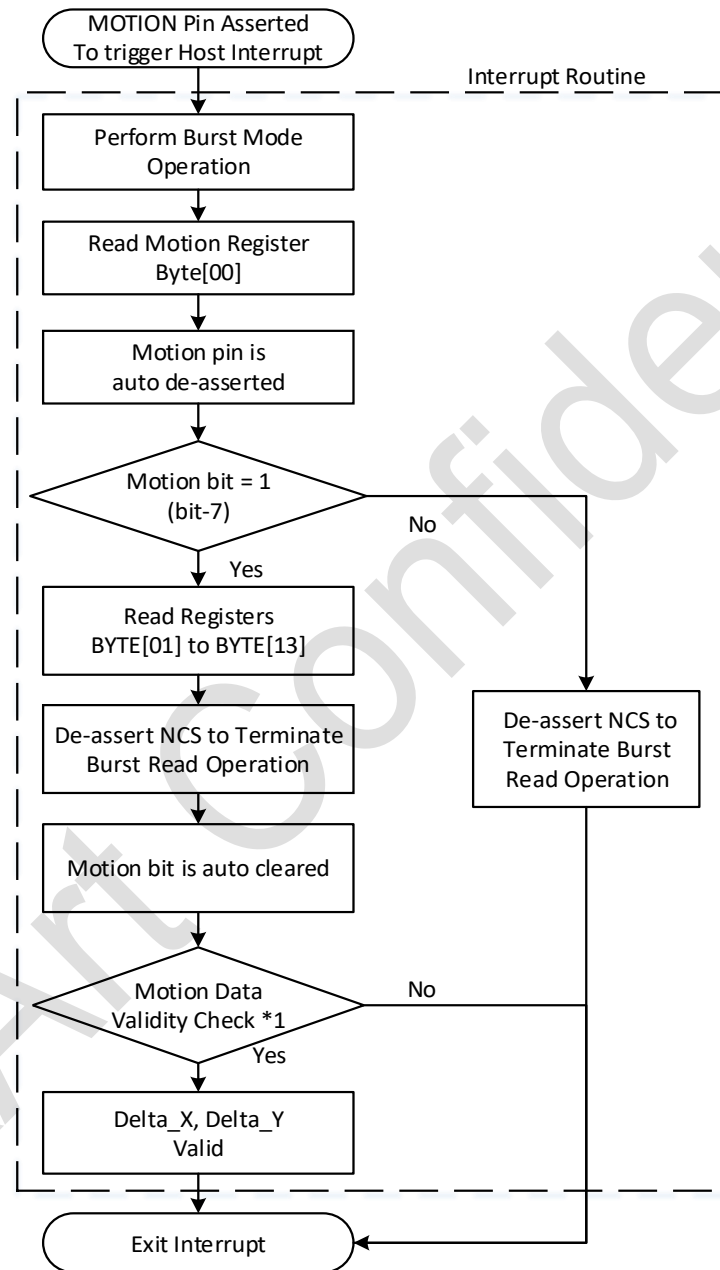
*2: Motion bit is auto cleared after reading *MOTION* register

*3: The polling interval is configured by the host

Figure 25. Polling Method – Burst Read

Interrupt Method

When MOTION pin is low, the register is pending host to read. Otherwise, the MOTION pin goes high when the register is cleared. When the chip detects the occurrence of motion, the MOTION pin is asserted and triggers the host to read data.



Note:

*1: Refer Section 7.4.3 for detail

Figure 26. Motion Data Access - Interrupt Method

7.4.3 Invalid Motion Data Condition

There is additional verification that required to check for validity of the motion data. The step to as follow:

1. Read *Observation* register bit-6 and bit-7 to check the current operation mode.
2. Access both *SQUAL* and *Shutter* register.
3. At the respective operation mode, to suppress false motion reports, discard Delta X and Delta Y values if the SQUAL and Shutter values meet the condition outlined in below table.

Mode	SQUAL	Shutter
0: Bright Mode	< 0x19	≥ 0x00FF80
1: Low Light Mode	< 0x46	≥ 0x00FF80
2: Super Low Light Mode	< 0x55	≥ 0x025998

7.4.4 Data Lost and Corruption

Motion data register storage limit is -32768 to 32767 for X-axis and Y-axis respectively. When motion is detected, motion data will be generated, accumulated and stored in registers. If host didn't read out motion data before data reach the limit, the data corruption may occur.

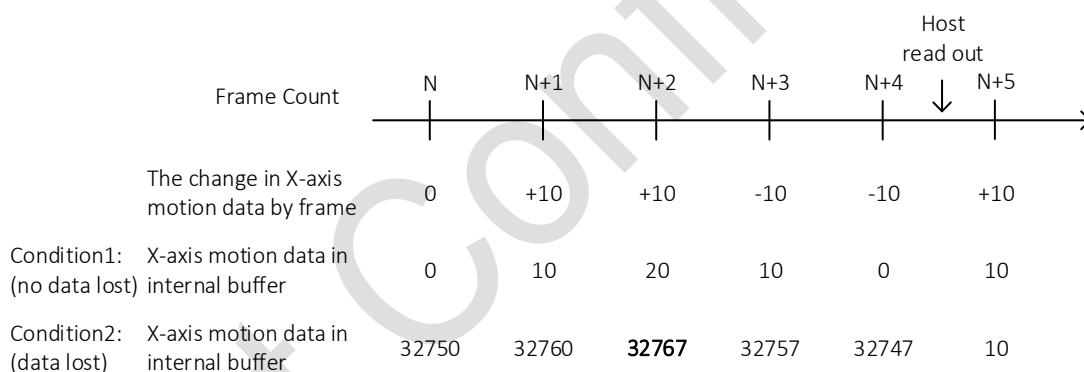


Figure 27. Example of Data Lost When Internal Buffer Reach limit

From frame N to frame N+4, the total change of motion data is 0.

- Condition1
If motion data register value is 0 in frame N, the host may read motion data value 0 in between frames N+4 and N+5.
- Condition2
If motion data register value is 32750 in frame N, the host may read motion data value 32747 in between frames N+4 and N+5.

The above two condition movement behavior are identical. However, condition 2 register hitting limit of 32767, host read value 32747 instead of 32750 where some motion data are lost.

As prevention,

1. The host requires to response to interrupt triggering as soon as possible for read motion data register.
2. The host need to shorten the polling interval.
3. The host require to monitor and avoid hitting motion data of -32768 to 32767 limit respectively.
4. The chip resolution can be optimized.

7.4.5 Raw Data Output

Raw Data Output is a special mode to download the full array of raw data using register read operation. This mode disables navigation and no other SPI activity is allowed during this period. To restore navigation for normal operation, software reset is required.

Power-on sequence should have been completed before performing Raw Data Output. Raw Data Output procedure is outlined below:

1. To enter Raw Data Output mode, perform the below register writes in sequence:
 - a. Write register 0x7F with value 0x00
 - b. Write register 0x67 with value 0x25
 - c. Write register 0x55 with value 0x20
 - d. Write register 0x7F with value 0x13
 - e. Write register 0x42 with value 0x01
 - f. Write register 0x7F with value 0x00
 - g. Write register 0x0F with value 0x11
 - h. Write register 0x0F with value 0x13
 - i. Write register 0x0F with value 0x11
2. Poll *RawData_Grab_Status* register (0x10) until getting bit 0 equal to 1.
3. Write register 0x13 with any value
4. Read register 0x13 for 1225 times to form a complete picture element array information.
5. Disable Raw Data output Mode, perform below register writes in sequence.
 - a. Write register 0x7F with value 0x00
 - b. Write register 0x55 with value 0x00
 - c. Write register 0x7F with value 0x13
 - d. Write register 0x42 with value 0x00
 - e. Write register 0x7F with value 0x00
 - f. Write register 0x67 with value 0xA5

6. To continue download the raw data for the next frame, repeat step 1 to step 5.
7. To resume navigation, issuing a software reset command to the chip and initialize the chip as per section 5.4.2.

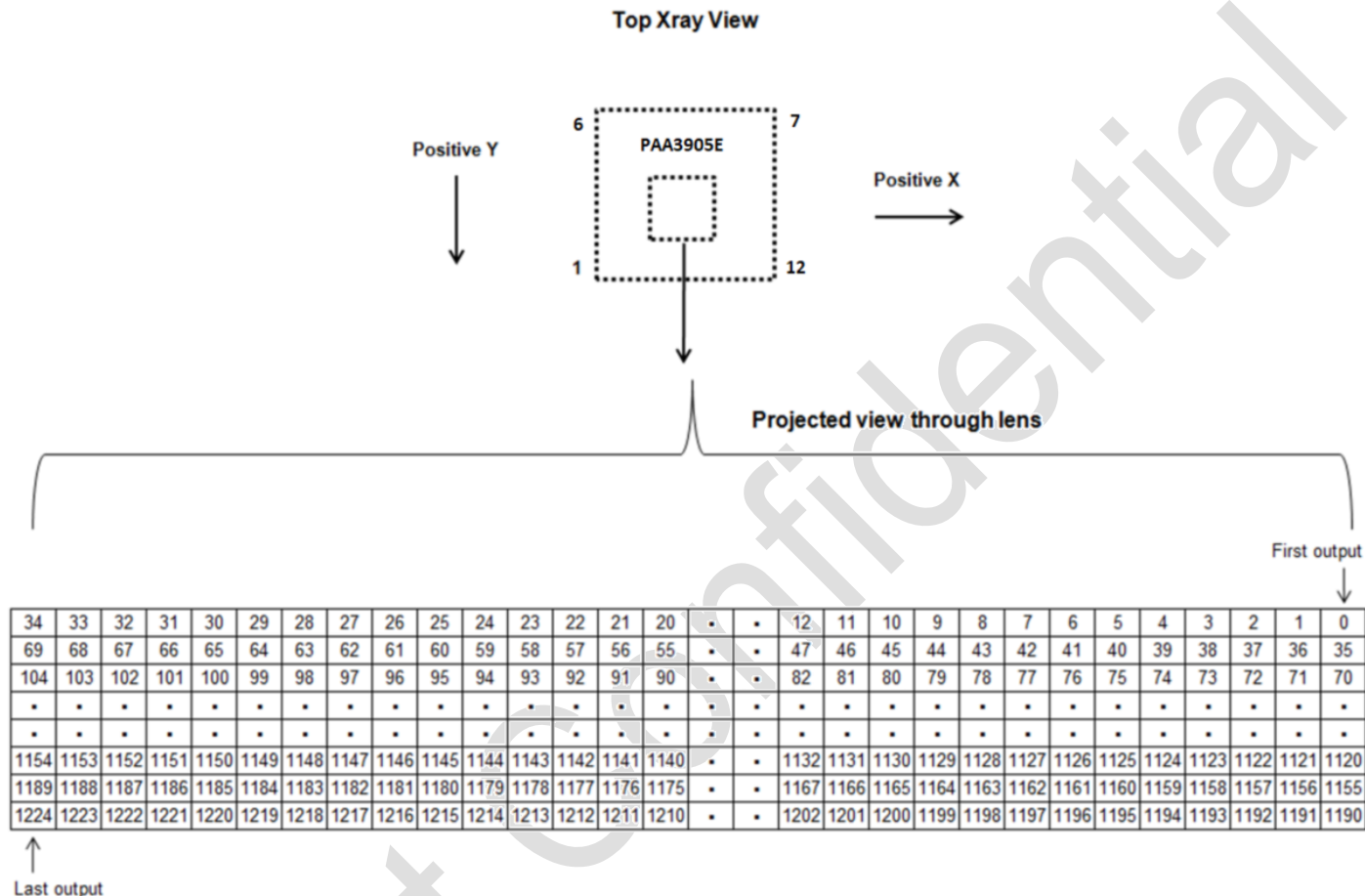
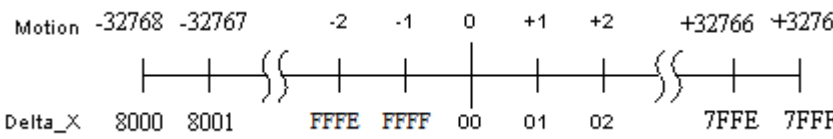


Figure 28. Raw Data Map

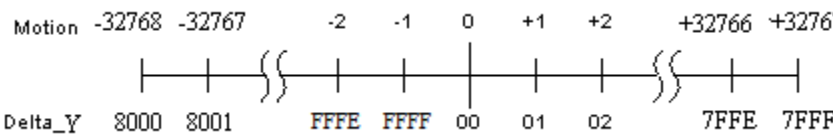
7.4.6 Related Registers

Usage	Name	Bank	Address
Output Access	<i>Motion</i>	-	0x02
	<i>Delta_X_L</i>	-	0x03
	<i>Delta_X_H</i>	-	0x04
	<i>Delta_Y_L</i>	-	0x05
	<i>Delta_Y_H</i>	-	0x06
Raw Data Output	<i>RawData_Grab_Status</i>	-	0x10
	<i>RawData_Grab</i>	-	0x13

Register Name	Motion														
Bank		Address	0x02												
Access	R	Default Value	0x00												
Description	This register allows user to determine if motion has occurred since the last time it was read. The procedure to read the motion registers (<i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i>) is as follows:														
	1. Read the <i>Motion</i> register. This will freeze the <i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> register values.														
	2. If the MOT bit is set, <i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> registers should be read in the given sequence to get the accumulated motion. Note that if <i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> registers are not read before the motion register is read for the second time, the data in <i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> registers will be lost.														
	3. To read a new set of motion data (<i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> registers), repeat from step (1).														
	Note:														
	1. To suppress false motion reports, discard Delta X and Delta Y values if the SQUAL and Shutter values meet the condition outlined in below table. Read <i>Observation</i> register bit-6 and bit-7 to check the current operation mode.														
	<table><tr><th>Mode</th><th>SQUAL</th><th>Shutter</th></tr><tr><td>0: Bright Mode</td><td>< 0x19</td><td>≥ 0x00FF80</td></tr><tr><td>1: Low Light Mode</td><td>< 0x46</td><td>≥ 0x00FF80</td></tr><tr><td>2: Super Low Light Mode</td><td>< 0x55</td><td>≥ 0x025998</td></tr></table>			Mode	SQUAL	Shutter	0: Bright Mode	< 0x19	≥ 0x00FF80	1: Low Light Mode	< 0x46	≥ 0x00FF80	2: Super Low Light Mode	< 0x55	≥ 0x025998
Mode	SQUAL	Shutter													
0: Bright Mode	< 0x19	≥ 0x00FF80													
1: Low Light Mode	< 0x46	≥ 0x00FF80													
2: Super Low Light Mode	< 0x55	≥ 0x025998													
	2. Writing anything to this register clears the <i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> registers.														
Bit Field	Name	Default Value	Description												
7	MOT[7]	0	Motion since last report 0: No motion 1: Motion occurred, data ready for reading in <i>Delta_X_L</i> , <i>Delta_X_H</i> , <i>Delta_Y_L</i> and <i>Delta_Y_H</i> registers.												
0	MOT[0]	0	0: Challenging surface is not detected 1: Challenging surface is detected												

Register Name	Delta_X_L		
Bank		Address	0x03
Access	R	Default Value	0x00
Description	16 bits 2's complement number. Lower 8 bits of <i>Delta_X</i>. X movement is in counts since last report. Absolute value is determined by resolution. 		

Register Name	Delta_X_H		
Bank		Address	0x04
Access	R	Default Value	0x00
Description	16 bits 2's complement number. Upper 8 bits of <i>Delta_X</i>. <i>Delta_X_H</i> must be read after <i>Delta_X_L</i> to have the full motion data. Note: It is recommended that registers 0x02, 0x03, 0x04, 0x05 and 0x06 be read sequentially.		

Register Name	Delta_Y_L		
Bank		Address	0x05
Access	R	Default Value	0x00
Description	16 bit 2's complement number. Lower 8 bits of <i>Delta_Y</i>. Y movement is in counts since last report. Absolute value is determined by resolution. 		

Register Name	Delta_Y_H		
Bank		Address	0x06
Access	R	Default Value	0x00
Description	16 bit 2's complement number. Upper 8 bits of <i>Delta_Y</i>. <i>Delta_Y_H</i> must be read after <i>Delta_Y_L</i> to have the full motion data. Note: It is recommended that registers 0x02, 0x03, 0x04, 0x05 and 0x06 to be read sequentially.		

Register Name	RawData_Grab_Status		
Bank		Address	0x10
Access	R	Default Value	0x00
Description	This register provides status of raw data grab process.		
Bit Field	Name	Default Value	Description
0	RDG[0]	0	0: Raw data is not ready. 1: Raw data is ready.

Register Name	RawData_Grab		
Bank		Address	0x13
Access	R/W	Default Value	0x00
Description	This register is used to read out the full array of raw data values, the chip needs to be held stationary for the duration of grabbing raw data until the full array is completely read out, as the information is read out one data at a time. This process is initialized by a single write of any value to this register (only need to be written once per frame). Reading this register will unload 8-bit raw data at a time.		

7.5 LED Control

The chip support external LED control to extend the appropriate illumination base on applications.

This section provides information and reference schematics to support an external LED circuitry. Synchronizing the LED_SYNC pulse control is a significant power saving feature for battery-powered applications.

NOTE: While this section aims to provide guidance in utilizing the LED_SYNC pin, user owns the responsibility to select the appropriate LED and design its circuitry to meet the desired end application.

For power saving purposes, LED control (LED_SYNC) is disable by default. Below is the step to enable the LED Control (LED_SYNC).

1. Power up the chip and initialize register settings as outlined in Section 7.1.1 Initial Flow.
2. Perform below register writes in sequence:
 - a. Write register 0x7F with value 0x14
 - b. Write register 0x6F with value 0x0C
 - c. Write register 0x7F with value 0x00

To monitor the LED_SYNC pulsing, connect the LED_SYNC pin with pull up 1kΩ resistor to V_{DDIO}. By probing the LED_SYNC pin, waveforms can be observed with the pulses as shown in below image.

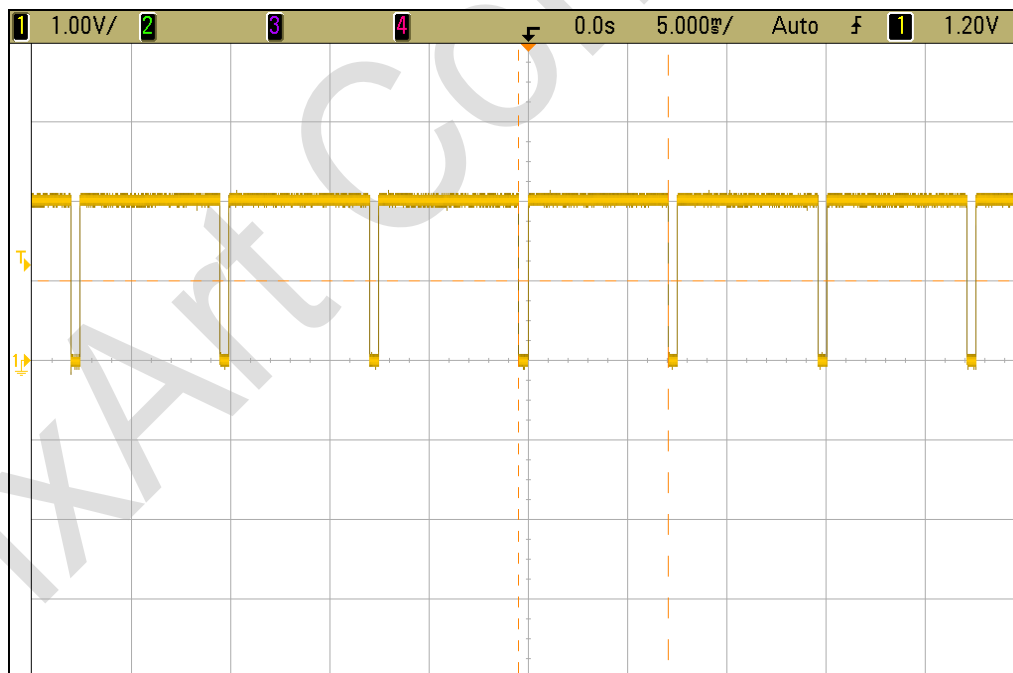


Figure 29. LED_SYNC pulsing

An example of synchronizing the LED_SYNC pulse to control an external LED circuitry using an IR LED is shown below:

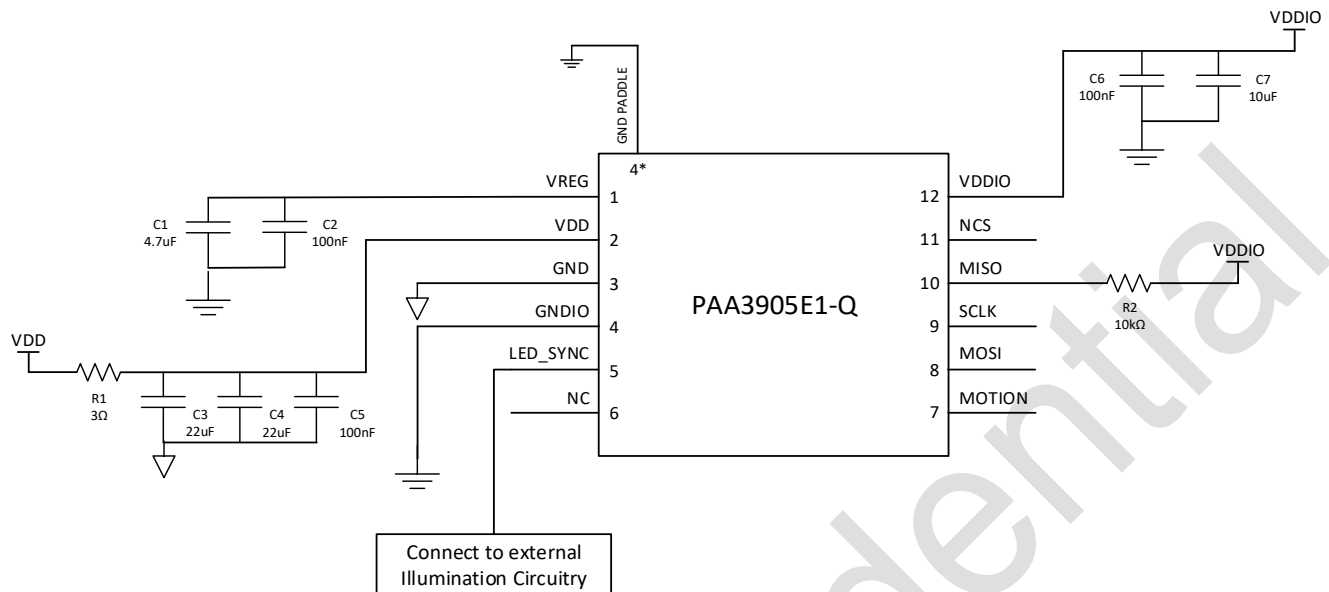


Figure 30. Reference schematics to drive external LED circuitry

7.6 Frame Synchronization (FS)

This chip supports frame synchronization function, a single or multiple chip can be synchronized to other devices at the desired frame rates through a series of register writes.

For example, in a drone application system, the output from the chip can be matched to the output rate of the IMU sensor to ensure data collection from these devices are from the same time stamp. In the case where multiple chip are employed in a system, synchronized data output from these multiple chips are crucial to provide accurate results in specific usage model.

7.6.1 Hardware Requirement

There is no additional hardware connection needed for Frame Synchronization (FS).

7.6.2 Single Chip Synchronization

The basis of this operation is to provide flexibility to host to start the frame of the chip at any given time. The procedure to synchronize the chip to the desired frame rate is as below:

1. Power-on the chip as per Section 7.1.
2. Perform below register writes in sequence. This issues a command to the chip to stop operation upon the completion of the current frame.
 - a. Write register 0x7F with value 0x07
 - b. Write register 0x41 with value 0x82
3. Delay for 100 ms to allow the chip sufficient time to complete the current frame's activities.
4. Read Register 0x15. Check Bit [7:6] for AMS mode and store it into a variable.

Bit [7:6]	AMS mode
0x0	Mode 0
0x1	Mode 1
0x2	Mode 2

5. Issue start operation command to the chip with the following register write.

Write register 0x1F with value 0x00

6. Start timer period before issuing start operation command to the chip. Use the timer period corresponding with chip's AMS mode as shown below:

Mode 0 & 1		Mode 2	
Timer (ms)	Frame Rate (fps)	Timer (ms)	Frame Rate (fps)
8.8	114	22.0	45

Note: Additional time (10% of actual AMS mode frame period) is added to buffer for clock variation between host and the chip.

7. Poll motion data through single read or burst read.

Note: Check *Motion* register for bit 6 to be zero (bit 6 = 0). If bit 6 is set (bit 6 = 1) for 3 consecutive frames. Exit the routine, power cycle the chip and repeat from Step 1.

8. Upon expiration of the timer period, repeat Step 4 to 7 to continue the synchronization routine and normal operation of the chip. If these steps are not repeated, there will be no data output from the chip.

9. To exit the synchronization routine, issue a soft reset command to the chip and configure the registers as per Section 7.1

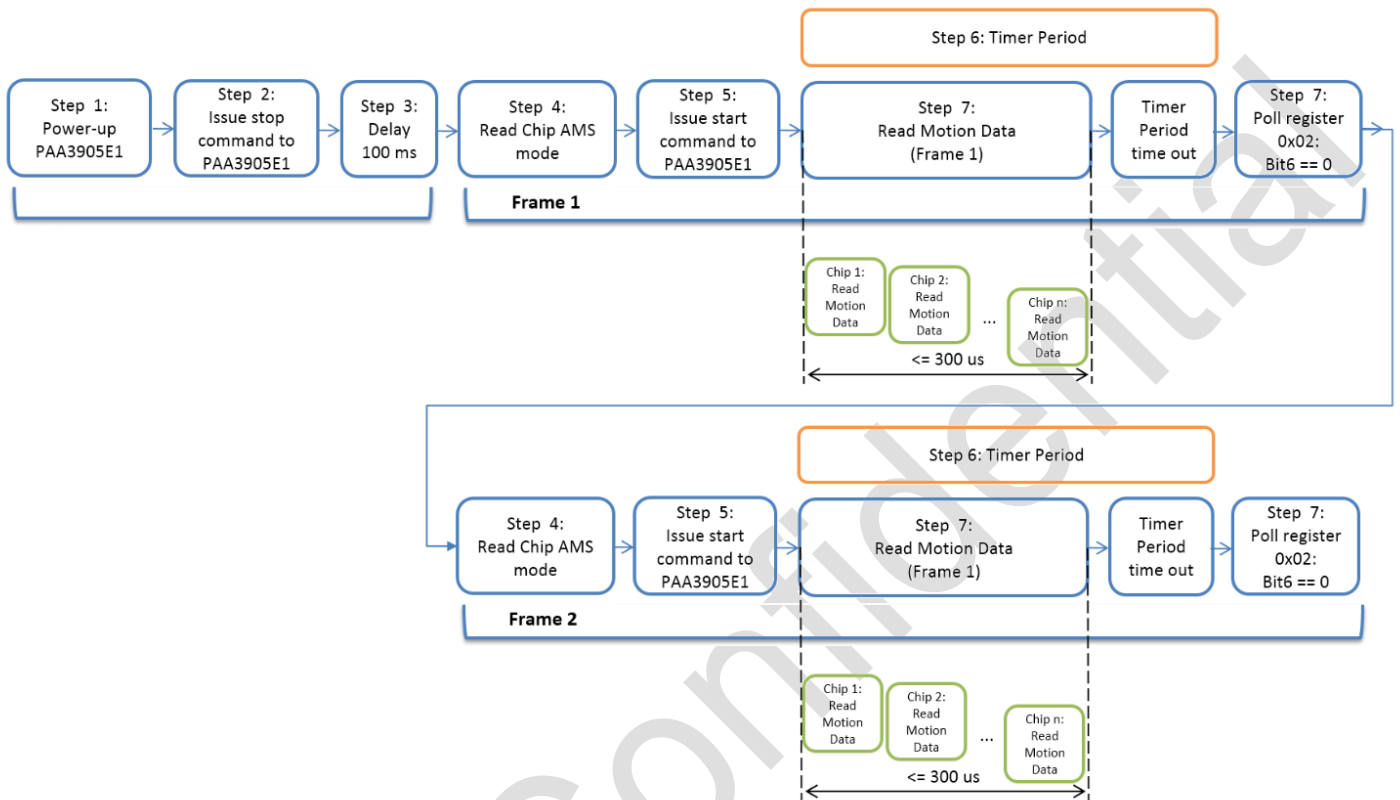


Figure 31. Single Chip Synchronization

7.6.3 Multiple Chips Synchronization

To synchronize multiple chips, additional steps are added to the steps in Section 7.6.2 Single Chip Synchronization. One chip will be selected to be a master chip with the rest being slave chips. Master chip is the only chip with AMS enabled. Slave chips will be following the same AMS mode as master sensor through host force AMS mode register write. All the chips will be synchronized by receiving the start frame register write at the same time. To reiterate, every chip's NCS pin should be active low during the start frame register write. This ensures all the chips start the frame at the same time as close as possible.

All motion data should be read immediately from one chip to another, in a sequential manner. The maximum period to read motion data for all the chips should not exceed 300 μ s (regardless of timer period used). This helps to ensure the motion data read from all the chips are synchronized. If motion data for the chips are read outside these time frames, the motion data obtained (for some of the chips) might not be from the same frame.

The procedure to synchronize multiple chips to the desired frame rate is as below:

1. Power-up the chip as per Section 7.1.
2. Perform below register writes in sequence to all master and slave chips at the same time. This issues a command to the chip to stop operation upon the completion of the current frame."
 - a. Write register 0x7F with value 0x07
 - b. Write register 0x41 with value 0x82
3. Delay for 100 ms to allow the chip sufficient time to complete the current frame's activities.
4. Read Register 0x15 from master chip. Check Bit [7:6] for AMS mode.

Bit [7:6]	AMS mode	Force_AMS_Mode
0x0	Mode 0	0x00
0x1	Mode 1	0x11
0x2	Mode 2	0x22

Perform below register writes in sequence to force AMS mode to all slave chips.

- a. Write register 0x7F with value 0x08
 - b. Write register 0x68 with value in Force_AMS_Mode
5. Issue start operation command to the chip with the following register write to all master and slave chips at the same time.

Write register 0x1F with 0x00

6. Start timer period before issuing start operation command to the chip. Use the timer period corresponding with the chip's AMS mode as shown below:

Mode 0 & 1		Mode 2	
Timer (ms)	Frame Rate (fps)	Timer (ms)	Frame Rate (fps)
8.8	114	22.0	45

Note:

Additional time (10% of actual AMS mode frame period) is added to buffer for clock variation between host and the chip.

7. Poll motion data through single read or burst read from one chip to another, starting with master chip followed by slave chips in a sequential manner

Note:

1. Check *Motion* register for bit 6 to be zero (bit 6 = 0) for all chips. If bit 6 is set (bit 6 = 1) for 3 consecutive frames for any of the chips, please exit the routine, power cycle the chip and repeat from Step 1.
 2. The maximum period to read motion data for all the chips should not exceed 300 μ s (regardless of timer period used)
8. Upon expiration of the timer period, repeat Step 4 to 7 to continue the synchronization routine and normal operation of the chip. If these steps are not repeated, there will be no data output from the chip.

9. To exit the synchronization routine, issue a soft reset command to the chip and configure the registers as per Section 7.1.

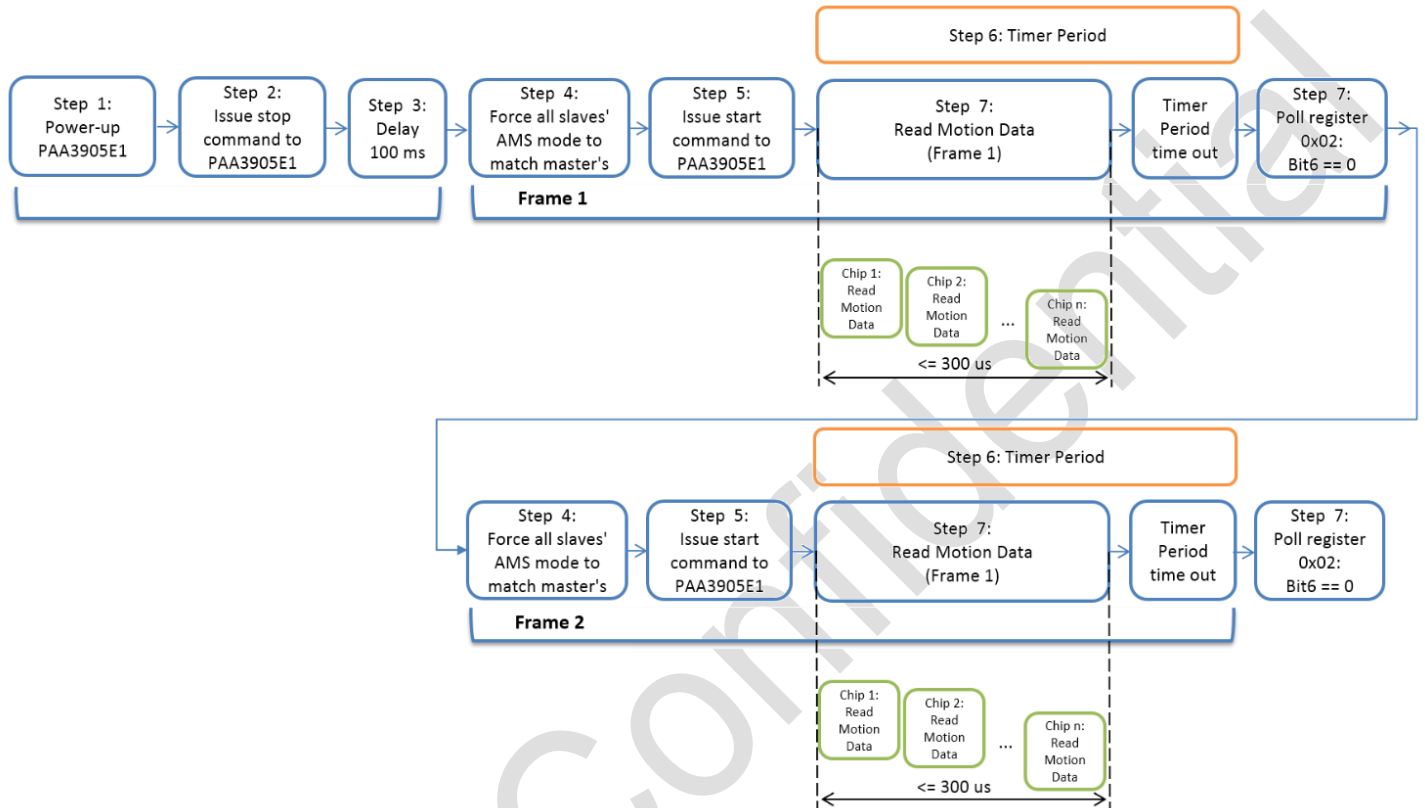


Figure 32. Multiple Chip Synchronization

8.0 Registers

8.1 Registers Map

Table 14. Register List

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x00	-	<i>Product_ID</i>	R	-	0xA2
0x01	-	<i>Revision_ID</i>	R	-	0x00
0x02	-	<i>Motion</i>	R/W	-	0x00
0x03	-	<i>Delta_X_L</i>	R	-	0x00
0x04	-	<i>Delta_X_H</i>	R	-	0x00
0x05	-	<i>Delta_Y_L</i>	R	-	0x00
0x06	-	<i>Delta_Y_H</i>	R	-	0x00
0x07	-	<i>SQUAL</i>	R	-	0x00
0x08	-	<i>RawData_Sum</i>	R	-	0x00
0x09	-	<i>Maximum_RawData</i>	R	-	0x00
0x0A	-	<i>Minimum_RawData</i>	R	-	0xFF
0x0B	-	<i>Shutter_Lower</i>	R	-	0x00
0x0C	-	<i>Shutter_Middle</i>	R	-	0x00
0x0D	-	<i>Shutter_Upper</i>	R	-	0x00
0x10	-	<i>RawData_Grab_Status</i>	R	-	0x00
0x13	-	<i>RawData_Grab</i>	R/W	-	0x00
0x15	-	<i>Observation</i>	R/W	-	0x00
0x16	-	<i>Motion_Burst</i>	R	-	0x00
0x3A	-	<i>Power_Up_Reset</i>	W	-	N/A
0x3B	-	<i>Shutdown</i>	W	-	N/A
0x4E	-	<i>Resolution</i>	R/W	-	0x2A
0x5B	-	<i>Orientation</i>	R/W	-	0xE0
0x5F	-	<i>Inverse_Product_ID</i>	R	-	0x5D

8.2 Register Description

8.2.1 Product ID

Register Name	Product_ID		
Bank		Address	0x00
Access	R	Default Value	0xA2
Bit Field	Name	Default Value	Description
7:0	Product_ID	0xA2	This value is a unique identification assigned to this model only. The value in this register does not change; it can be used to verify that the serial communications link is functional.

Register Name	Revision_ID		
Bank		Address	0x01
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
7:0	Revision_ID	0x00	This register contains the current IC revision. It is subject to change when new IC versions are released.

Register Name	Inverse_Product_ID		
Bank		Address	0x5F
Access	R	Default Value	0x5D
Bit Field	Name	Default Value	Description
7:0	Inverse_Product_ID	0x5D	This value is the inverse of the <i>Product_ID</i> . It is used to test the SPI port hardware.

8.2.2 Operational Control Related Register

Register Name	Resolution		
Bank		Address	0x4E
Access	R/W	Default Value	0x2A
Bit Field	Name	Default Value	Description
7:0	Resolution	0x2A	This register sets the X and Y resolution of the chip. To calculate the approximate resolution value of each register setting, use the formula below: Approximate Resolution = (Register Value + 1) x (200 ÷ 8600), each register step is about 2.3% from the default value in Figure 3. Resolution versus Height Chart. The maximum register value is 0xFF (6x of default). The minimum register value is 0.

Register Name	Orientation		
Bank		Address	0x5B
Access	R	Default Value	0xE0
Description	This register sets the orientation of the reported X and Y positions. This includes swapping of X and Y; and flipping (inverting) the direction of X and Y axis. If both are selected, swapping is done before flipping.		
Bit Field	Name	Default Value	Description
7	ORT ₇	1	Invert X direction 0: Not inverted 1: Inverted
6	ORT ₆	1	Invert Y direction 0: Not inverted 1: Inverted
5	ORT ₅	1	Swap X and Y 0: No Swap 1: Swap

8.2.3 Operational check Related Registers

Register Name	SQUAL		
Bank		Address	0x07
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
7:0	SQUAL	0x00	The <i>SQUAL</i> (Surface quality) register is a measure of the number of valid features visible by the chip in the current frame. Use the following formula to find the total number of valid features: Number of Features = <i>SQUAL</i> Register Value x 4 The maximum <i>SQUAL</i> register value is 0xFF. Since small changes in the current frame can result in changes in <i>SQUAL</i>, variations in <i>SQUAL</i> when looking at a surface are expected. <i>SQUAL</i> values are only valid in run mode.

Register Name	RawData_Sum		
Bank		Address	0x08
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
7:0	RawData_Sum	0x00	This register is used to find the average raw data value. To find the average raw data value, use the formula below: Average Raw Data = (Register Value x 8192) ÷ 1225 The maximum register value is 0x98. The minimum register value is 0. The <i>RawData_Sum</i> value can change every frame.

Register Name	Maximum_RawData		
Bank		Address	0x09
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
7:0	Maximum_RawData	0x00	Maximum raw data value in current frame. Minimum value = 0, maximum value = 255. The maximum raw data value can change every frame.

Register Name	Minimum_RawData		
Bank		Address	0x0A
Access	R	Default Value	0xFF
Bit Field	Name	Default Value	Description
7:0	Minimum_RawData	0xFF	Minimum raw data value in current frame. Minimum value = 0, maximum value = 255. The minimum raw data value can change every frame.

Register Name	Shutter_Lower		
Bank		Address	0x0B
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
7:0	Shutter_Lower	0x00	Lower byte of the 23-bit <i>Shutter</i> register.

Register Name	Shutter_Middle		
Bank		Address	0x0C
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
7:0	Shutter_Middle	0x00	Middle byte of the 23-bit <i>Shutter</i> register.

Register Name	Shutter_Upper		
Bank		Address	0x0D
Access	R	Default Value	0x00
Bit Field	Name	Default Value	Description
6:0	S[22:16]	0	Upper 7-bit of the 23-bit <i>Shutter</i> register. Unit is clock cycles of the internal oscillator / 4. Read <i>Shutter_Upper</i> first, follow by <i>Shutter_Middle</i> and then <i>Shutter_Lower</i> . They should be read consecutively. The shutter is adjusted to keep the average raw data values within normal operating range. The shutter value is checked and automatically adjusted to a new value if needed on every frame when operating in default mode.

8.2.4 Troubleshooting Related Registers

Register Name	Observation		
Bank		Address	0x15
Access	R/W	Default Value	0x00
Description	User must clear the register by writing 0x00, wait for 30 ms, and read the register. The active processes <i>OB</i> [5:0] will have set their corresponding bits. The read back value for <i>OB</i> [5:0] should be 0x3F. <i>OB</i> [7:6] indicates the current operation mode of the chip.		
Bit Field	Name	Default Value	Description
7:6	<i>OB</i> [7:6]	0	0: Bright Mode 1: Low Light Mode 2: Super Low Light Mode 3: Reserved
5:0	<i>OB</i> [5:0]	0	0x3F: chip is working correctly Other value: recommend to issue a software reset command to the chip and configure the register as per section 7.1 in this datasheet.

Revision History

Revision	Date	Description
0.8	04 Sept 2020	Initial Release